the discovery of the positron. The main advantage
of the elementary particles produced in cosmic rays is that they have enormous energies
(and of course free). These energies are far-greater than what could be produced in the laboratories even today. The major disadvantages are that flux of cosmic rays is extremely small and their energies are completely uncontrolled. For more details on cosmic rays, refer to Chapter 9.

8.2.2 Particle Accelerators

As discussed in Chapter 6, particle accelerators accelerate charged particles, like electrons and protons, to very high energy and bombard them on some suitable targets, like hydrogen. These charged particles may also collide head-on with each other. Depending upon the energy of the accelerated charged particles, the resulting reaction debris contain particles like pions, kaons, muons, and even antiparticles like antiproton, etc. With accelerators it is possible to generate most intense beams of the elementary particles, and by proper choice of magnets and absorbers, beam of desired elementary particles can be extracted. The major disadvantage of the accelerator produced elementary particles is the limit on the energy available which depends on the design of the accelerator. This also limits the number of particles produced. Further, high-energy accelerators are extremely costly.

8.2.3 Nuclear Reactors

When heavy nuclei like ^{235}U undergo fission in the nuclear reactors, they emit variety of elementary particles like neutrons (n), neutrinos (n), b-particles (e^+, e^-) and g-rays. Nuclear reactors thus are intense source of these elementary particles.

8.3 TYPES OF INTERACTIONS

In order to understand the behaviour of the elementary particles, i.e. their formation and decay, one needs to understand different types of fundamental interactions[2] among them. These interactions which fall into the following four categories:

[2] One also hears about many other interactions like chemical, electrical, mechanical, atomic forces, but all such interactions belong to the above-mentioned interactions.

1. Gravitational interactions.
2. Electromagnetic interactions.
3. Strong interactions.
4. Weak interactions.

In fact, all the known processes occurring in nature (starting from subnuclear to extra-galactic levels) can be understood as manifestations of these four interactions. Note that in particle physics, the words force and interaction are used for the same purpose. Now we give brief introduction of these interactions.

8.3.1 Gravitational Interactions

This was the first interaction known to humans. First law of gravitation was formulated in 1666 by Newton. The gravitational force is mainly responsible for keeping heavenly bodies like moons, planets, stars, etc. in their orbits. This force depends upon the masses of the two objects and the distance between them. It is always attractive and its range is infinite. In order to compare the relative strength of different interactions, we define a dimensionless coupling constant as

$$g_m = G_N \frac{m^2}{\hbar c}$$

where G_N is gravitational constant = 6.66 □ 10^{-11} Nm^2/kg^2, m is the mass of the objects between which gravitational interaction is operating, □ = $\frac{h}{2p}$, h is the Planck's constant and c is the speed of light. The dimensionless coupling constant between two nucleons can be calculated by taking m □ 1.67 □ 10^{-27} kg

$$g_m = \frac{2 \times p \times 6.66 \times 10^{-11} \times (1.67 \times 10^{-27})^2}{6.626 \times 10^{-34} \times 3 \times 10^8} \quad \square\ 6 \square\ 10^{-39}$$

This interaction is universal, i.e. every particle, whether massive or massless, is subject to the gravitational force. The carrier of gravitational

interactions is conjectured to be a graviton. The mass of graviton is zero and thus it travels with velocity which is equal to that of the light. This interaction is extremely small for elementary particles, and hence is usually neglected.

8.3.2 Electromagnetic Interactions

The theory of electromagnetism was formulated on the great experimental works, started with appearance of Coulomb law of electrostatics in 1776. Later due to the works of Faraday and Maxwell, a proper field theory of electromagnetic forces was finalized around 1860. The electromagnetic interaction provides a force between two charges, which is repulsive, if the two charges are similar (i.e. both positive or both negative) and attractive, if the charges are opposite (i.e. one is positive and other is negative). All the chemical and biological reactions (*including life*) are basically controlled by the electromagnetic interactions. This interaction is also responsible for formation of electron–positron pair from *g*-rays, and vice versa. This interaction also plays a role in nuclear structure through the mutual repulsion between protons. It causes several electromagnetic decays like decay of neutral pion into two photons. The dimensionless coupling constant for electromagnetic interactions is given by

$$a = \frac{e^2}{4p\epsilon_0 \times \hbar c}$$

$$= \frac{2p \times (1.602 \times 10^{-19})^2}{4p \times 8.854 \times 10^{-12} \times 6.626 \times 10^{-34} \times 3 \times 10^8}$$

$$= \frac{1}{137}$$

The characteristic interaction time of electromagnetic interactions is $\sim 10^{-16}$ s. The carrier of this interaction is a photon with rest mass zero and the range of this interaction is infinite.

Until 1930s, only the above two forces were known. Around 1934, it was realized that there are other kinds of natural forces, which were called strong and weak forces.

8.3.3 Strong Interactions

Strong interactions hold several nucleons together in a nucleus against the electromagnetic repulsion of the protons. Strong force is charge independent, i.e. it is the same between *p*–*p*, *n*–*n* and *p*–*n*. The range of the strong interactions is about 10^{-15} m and the interaction time is about 10^{-23} s. This is roughly the time taken by light to travel across a proton. In 1935, Yukawa explained this force at nuclear level through the exchange of particles called *pions*. However, the dimensionless pion-nucleon coupling constant representing the strength of strong interactions is about $\frac{g^2_{NNp}}{\hbar c} = 14,$ indicating that the strength of electromagnetic interactions is about a one thousandth part (10^{-3}) of the strong interactions. Strong interactions are responsible for *a*-decay, nuclear reactions, production of hadrons, etc. In fact, strong interactions occur at the quark level through exchange of so-called *gluons* carrying spin 1 and massless, and nuclear forces are expected to be residual forces arising out of the quark level interactions, similar to the van der Waals forces in electromagnetism.

8.3.4 Weak Interactions

Unlike other interactions, weak interactions are unique in that they do not form any bound system, and only show through decay of the elementary particles. For instance, weak interaction is always responsible for all radioactive decays in which *b*-particles (e^- or e^+) are involved. The range of weak interactions is very small $\sim 10^{-18}$ m, which is about 0.1% of the diameter of a proton. The dimensionless coupling constant for weak interactions is $g_W \sim 10^{-13}$ and the interaction time $\sim 10^{-10}$ s. In 1934, Fermi developed the first theory of weak interactions known as the *Fermi theory of beta decay* with coupling $G_F = 1.166 \times 10^{-5}$ GeV^{-2}. Later, the weak interactions were thought to be manifestation of the exchange of certain spin one particles. Final theory of weak interactions, known as the Glashow, Weinberg and Salam theory in 1967, involves intermediate vector *bosons*, $W^{\pm}$, Z^0, as the carrier of weak interactions. They have masses

greater than 80 GeV and spin = 1. A summary of various interactions is shown in Table 8.1.

TABLE 8.1 Basic properties of fundamental interactions

Interaction	Strength	Interaction time (s)	Range (m)	Quanta of force	Spin
Strong	1	10^{-23}	10^{-15}	*p*-mesons, gluons	0
					1
Electromagnetic	$\sim 10^{-3}$	10^{-16}	Infinite	Photon	1
Weak	10^{-13}	10^{-10}	10^{-18}	*W*, *Z* bosons	1
Gravitational	10^{-39}	–	Infinite	Graviton[3]	2

[3]

[3] Graviton is a hypothetical particle, whose existence is yet to be confirmed experimentally.

8.4 CLASSIFICATION OF ELEMENTARY PARTICLES

In cosmic rays and high-energy accelerator experiments, a large variety of elementary particles were discovered. The total number of elementary particles around 1957 was about 30 and by 1965, this number rose to about 200 and now it has become over 500. As the number kept on growing further, so need was felt to classify these particles into different groups. All fundamental forces can act on the elementary particles at the same time. Since they have differences in their strengths, interaction time and range, etc., these parameters can be used to classify the elementary particles. The most common classification is as shown in Figure 8.1, in which the elementary particles are divided into two main categories:

- Quanta of forces.
- Matter particles.

Matter particles are classified into the following two categories:

- Leptons.
- Hadrons.

The hadron family further consists of baryons and mesons. This classification has some similarities with atomic structure. All the leptons are

electron-like, quanta are photon-like, whereas the baryons are nucleon-like, and mesons are pion-like in nature.

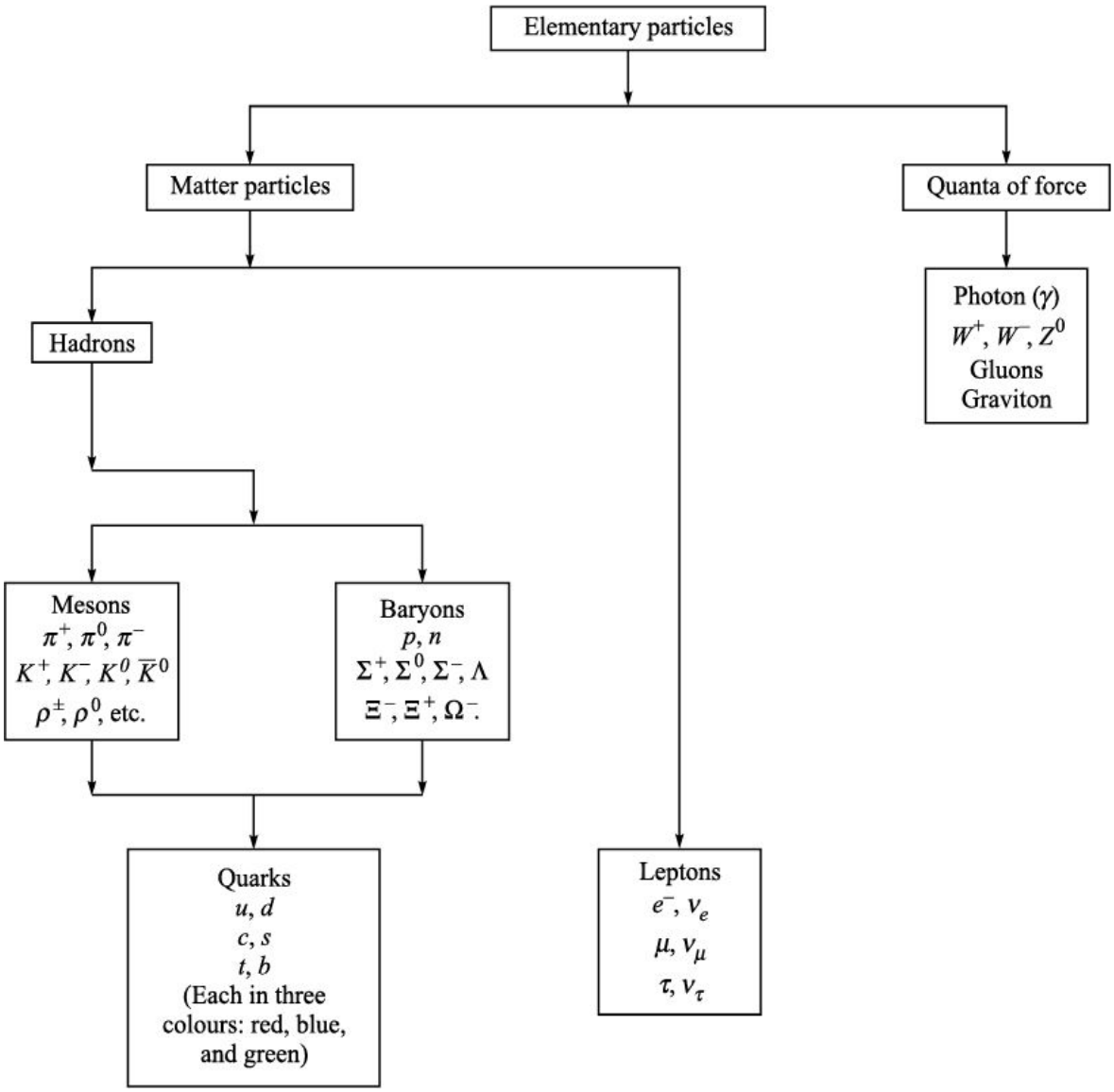


Figure 8.1 Classification of elementary particles.

8.4.1 Quanta of Forces

Corresponding to four different interactions, strong, weak, electromagnetic and gravitational, there are certain quanta of the forces which act as the mediator of these interactions. Their properties in brief are given in Table 8.2. *W*- and *Z*-bosons, having respective decay widths of
2.1 GeV and 2.5 GeV, decay to lepton pairs or hadrons. Decay width (□) of any unstable particle is related to its lifetime (*t*) as

$$\square = \square/t$$

Photon and gluons are electrically neutral and massless. These are treated as stable particles. There are eight kinds of gluons differing in their colour quantum number.

TABLE 8.2 Quanta of forces

Quanta	Interaction	Spin	Mass
Gluons	Strong	1	0
W^+ and W^- bosons	Weak	1	80.4 GeV
Z boson	Weak	1	91.2 GeV
Photon	Electromagnetic	1	0
Graviton	Gravity	2	0

8.4.2 Matter Particles

Leptons do not participate in the strong interactions; neutral leptons participate only in the weak interactions, whereas the charged leptons participate in electromagnetic as well as weak interactions. Hadrons participate in all the interactions. Later we shall learn that all the hadrons, having size of the order femtometer, are in fact composed of quarks and/or antiquarks. Remember that gravitational interaction is universal, but does not play a significant role at the present energy scales (Table 8.3).

TABLE 8.3 Matter particles

Particles	Strong	Electromagnetic	Weak
Hadrons	Yes	Yes	Yes
Charged leptons	No	Yes	Yes
Neutral leptons	No	No	Yes

8.4.3 Antiparticles

Scientific history of antiparticles (and of antimatter) started in 1928 when Nobel laureate physicist P.A.M. Dirac dramatically predicted on theoretical basis the existence of a particle that in certain respects was similar and in other respects in direct contrast with an electron. After the successes of nonrelativistic quantum mechanics, it was natural to perform its relativistic generalization, which faced certain difficulties. To solve them, Dirac constructed a relativistic equation, now known as the *Dirac equation*, which has four solutions. First two solutions were used to describe electron (e^-),

the last two solutions, having negative energies, were interpreted as antiparticle to the electron. Imagine the amazement when in 1932 Carl Anderson observed this antiparticle in a cloud chamber. Later, it was christened positron which has the same mass as that of the electron but opposite charge (e^+). The discovery of the positron was an important event for two reasons. Firstly, it marked the beginning of a series of discoveries of new elementary particles in the cosmic radiation made in the next two decades. Secondly, it provided the first experimental proof of relativistic quantum theory. There has been no looking back on the story of antiparticles since then, as the idea of antiparticle is not limited to electrons only. In later experiments, pions and muons were found to come with both particle and antiparticle species. In 1957, *CPT* (*C* = charge conjugation, *P* = parity, *T* = time reversal) theorem was formulated
(see Section 8.7.10) which enunciates that "*for every particle that exists in nature there is a corresponding antiparticle*". The theorem also predicts that the antiparticle has the same mass as its particle, the same lifetime (if the particle is not stable), the opposite charge and the opposite of all other internal quantum numbers. Certain particles act as their own antiparticle, like photon and p^0.

8.4.4 Observation and Production of Antimatter

Now a question arises. How to create antiparticles? A pair of a particle and its antiparticle can be created from energy in the form of electromagnetic radiation. This phenomenon is known as *pair creation*. In fact, in the Anderson experiment, high-energy cosmic rays hit the lead piece in the cloud chamber and created electron–positron pairs, and the two particles curved in opposite directions in the applied magnetic field. Reverse of this process, known as *pair annihilation* is also possible. A particle and its antiparticle upon coming in contact with
each other annihilate yielding energy in the form of radiation. The phenomena of pair production and particle–antiparticle annihilation prove conclusively the equivalence[4] and interconvertibility of mass and energy.

4. Incidentally, this equivalence was predicted by Albert Einstein much earlier (1905) in his theory of relativity. He demonstrated that energy, (E) momentum $(\vec{p})$ and mass (m) of a free particle are

related as $E^2 = p^2c^2 + m^2c^4$. Thus, a particle at rest carries a certain amount of energy ($E = mc^2$) called rest mass energy.

Today, antiparticles of all the basic constituents of matter, such as leptons and quarks, and quanta of forces have been observed experimentally. Also antiproton ( ), antineutron ( ) and a variety of others in this category (hadrons) also have been observed. In 1955, an accelerator Bevatron was built at Berkeley which gave just enough energy to create antiprotons. This was achieved by accelerating protons (p) to sufficient energy (6.2 GeV), colliding them with nuclei to generate the following process:

$p + p \square 3p +$

This was followed by the discovery of antineutron $(\overline{n})$ in 1956 and anti-$\square$ hyperon in 1958. Then there are instances in radioactive substances where an antiparticle, like positron or antineutrino, is emitted by the nucleus which is made up of matter. In recent years researchers have attempted to produce atoms (or better term is antiatoms) made up of antiparticles only. Simplest antiatom[5] should be *antihydrogen* formed by a positron orbiting around an antiproton. Experiments, using antiproton decelerator and plasma bottles to collect antiprotons and positrons, are in progress which aim to synthesize antihydrogen at sub-kelvin temperatures.

5. Another system of interest could be atomic protonium, a bound state of proton and antiproton. It is similar to a hydrogen atom, except that its electron is replaced by antiproton. However, this system is highly unstable, due to rapid p annihilation.

8.5 MASS SPECTRA AND DECAYS OF ELEMENTARY PARTICLES

Except for a few particles, like electrons, neutrinos, and protons, almost all the particles are unstable, and decay to other lighter particles through one or more of these interactions. Every particle is denoted by a special symbol, and its electric charge is usually shown as superscript. Besides the charge, the particles carry unique properties, like mass, magnetic moments, and other quantum numbers. Unstable particles may have several decay modes.

8.5.1 Leptons

Electron, the first lepton, was discovered in 1897. In 1937 Anderson and others found a particle with a mass of about 105 MeV, which was named

muon[6]. The experiments on the energy spread of the electrons from nuclear *b*-decay, made in the 1920s, indicated an apparent failure of the energy and momentum conservation laws. Later work showed that the conservation of angular momentum was also violated. In 1933, W. Pauli suggested that these conservation laws could be saved by postulating the existence of a spin 1/2 $\hbar$ particle of zero (or nearly zero) mass and no electrical charge, which was later called neutrino. In 1962, it was recognized that there are two kinds of neutrinos. Until 1975, four leptons[7] were known: electron, muon, electron-neutrino and muon-neutrino. In 1975, a fifth lepton named tau was discovered by Perl and his collaborators. This lepton has a mass about 3500 times the mass of the electron, that means even greater than that of the proton. Tau-lepton also has its corresponding neutrino partner, called tau-neutrino. So now the total number of leptons is six, three charged and three neutral: e^-, m^-, t^-, n_e, nm and nt; and of course their antiparticles. Leptons do not have any internal structure at the present energy scales, and all of them have spin 1/2. Range of their masses lies from zero to 1.78 GeV. Basic properties of the leptons and antileptons are given in Tables 8.4 and 8.5. Some of their special features are discussed below.

6. In the beginning this was also called *m*-mesons, but now the word meson has been reserved for different type of particles, i.e. hadrons.

7 Note the nomenclature, lepton, meson and baryon, has only historical relevance. For instance, originally, leptons (meaning *light: Greek*) were expected to have small masses. Similarly, mesons (meaning *intermediate*) were though to have intermediate masses, smaller than that of the baryons (meaning *heavy*). However, now the tau-lepton and many meson (D and B) states, heavier than the proton and even other baryons, have been observed in the high-energy accelerator experiments.

TABLE 8.4 Summary of lepton properties

Particle	Charge of particle	Mass (MeV)	Spin (□)	Lifetime	Decay modes
e^-	–1	0.511		Stable	Nil
n_e	0	<2 eV		Stable	Nil
m^-	–1	105.7		$2.2 \square 10^{-6}$ s	$m^- \square e^- + nm +$.
nm	0	<0.19		Stable	Nil
t^-	–1	1777		$0.29 \square 10^{-12}$ s	$t^- \square m^- +$. $+ nt$
					$t^- \square K^- + nt$
nt	0	<18.2		Stable	Nil

TABLE 8.5 Summary of antilepton properties

(Mass, spin and lifetime of antileptons are identically the same as that of the corresponding lepton partner)

Antiparticle	Charge of antiparticle	Decay modes
e^+	+1	Nil
	0	Nil
m^+	+1	$m^+ \rightarrow e^+ + \ldots + n_e$
	0	Nil
t^+	+1	$t^+ \rightarrow m^+ + nm + \ldots$
		$t^+ \rightarrow K^+ + \ldots$
	0	Nil

Electron (e^-)

(i) This particle is negatively charged. e^--electron and its antiparticle (e^+) is known as the positron.

(ii) Electrons e^- are generally produced in two ways:

(a) In metals there are lot of free e^-, which can be ejected from the surface of metal simply by heating the metal piece to high temperature. They can also be produced by ionizing the gases and extracting e^-, using suitable electric or magnetic fields.

(b) In radioactivity, e^- are emitted when a neutron decays to a proton

$$n \rightarrow p + e^- + \bar{n}_e$$

The source of e^+ is also radioactive decay in which a proton (inside nucleus) decays to a neutron

$$p \rightarrow n + e^+ + n_e$$

or by electron–positron pair production by gamma rays

$$g \rightarrow e^+ + e^-$$

(iii) Mass of both electron and positron is 0.511 MeV.

(iv) Magnetic moment of electron is one Bohr magneton, $m_e = e\hbar/2m_e c$.

(v) e^- is stable particle. However, when an e^+ meets e^-, both of them annihilate emitting photons as

$$e^+ + e^- \rightarrow g + g$$

Muon (m^-)

(i) After positron, muon was the next particle observed in the cosmic ray experiments in 1936. Like electron, this particle occurs as negatively charged m^- and its antiparticle is m^+.

(ii) m-mesons are generally produced in the pions decays:

$$p^- \rightarrow m^- + \bar{\nu}_m$$

$$p^+ \rightarrow m^+ + \nu m$$

(iii) Mass of both muons $m^{\pm}$ is 105.7 MeV.

(iv) Magnetic moment of muon is $\mu m = e\hbar/2mmc$.

(v) Muons $m^{\pm}$ decay as under:

$$m^- \rightarrow e^- + \bar{\nu}_e + \nu m$$

$$m^+ \rightarrow e^+ + \nu_e + \bar{\nu}_m$$

(vi) Lifetime of $m^{\pm}$ is 2.2×10^{-6} s.

Tau-Lepton (t^-)

(i) This particle occurs as negatively charged t^- and its antiparticle is t^+. These particles are originally observed in the electron–positron collision at very high energies, as

$$e^+ + e^- \rightarrow t^+ + t^-$$

followed by decays of the tau-leptons.

(ii) Mass of both $t^{\pm}$ is 1.777 GeV.

(iii) Like muon, $t^{\pm}$ can also decay through pure leptonic modes

$$t^- \rightarrow m^- + \bar{\nu}_m + \nu t$$

$$t^+ \rightarrow m^+ + \nu m + \bar{\nu}_t$$

however, being massive they have several decay modes. Their dominant modes emit one or more hadrons. For instance,

$$t^{-} \rightarrow K^{-} + nt$$

(iv) Lifetime of $t^{\rightarrow}$ is much smaller than that of the muon, i.e., 0.29×10^{-12} s.

Neutrinos

Like charged leptons, neutrinos have antipartners called *antineutrinos*, which have been observed experimentally. All the neutrinos (and antineutrinos) are electrically neutral, massless[8] and have spin ½ ħ. They participate in the weak interactions only.

8. Neutrinos were originally thought to be massless; however, some recent experiments on solar-neutrinos and neutrino oscillations observed in super kamiokande detector in Japan in 1998 endow some tiny special masses with them.

Electron-Neutrino (n_e)

(i) n_e and $\bar{n}_e$ are produced when in a nucleus a proton or a neutron decays as:

$$p \rightarrow n + e^{+} + n_e$$

$$n \rightarrow p + e^{-} + \bar{n}_e$$

It may be noted that proton can convert to neutrons only when inside a nucleus.

(ii) Following neutrino-nucleon reactions take place in nature:

$$n_e + n \rightarrow p + e^{-}$$

$$\bar{n}_e + p \rightarrow n + e^{+}$$

However, the cross-section for these reactions is extremely small ~1.2×10^{-43} cm^2.

(iii) Experimental upper limit on its mass is 2 eV based on tritium beta decay.

In order to detect this small cross-section, very large amount of target material and high flux of neutrinos were required. Finally, the existence of neutrino (in fact antineutrino) was established in nuclear reactor experiment by Reines and Cowan in 1959. (See Sections 3.6.6 and 3.6.7.)

Muon-Neutrino (*nm*)

(i) Since neutrinos are also emitted along with muons in the decays of pions and kaons, a question of considerable significance was asked: Are the neutrinos associated with muons necessarily the same as those emitted in nuclear *b*-decay? In 1962, experimental evidence showed that they are different, and thus muon-neutrino was identified, when it was realized that the reaction

$$vm + n \rightarrow m^{-} + p,$$

performed with neutrinos beam obtained from the pion or kaon decays, always takes place but never the following reaction:

$$nm + n \rightarrow e^{-} + p$$

In this experiment, muon-neutrinos were taken from the pion decay. Muon-neutrinos are thus different from the electron-neutrinos.

(ii) *nm* and $\bar{n}_m$ are produced in the decay of *p*-mesons as under:

$$p^{+} \rightarrow m^{+} + nm$$

$$p^{-} \rightarrow m^{-} + \bar{n}_m$$

(iii) Experimental upper limit on their mass is 0.19 MeV.

Tau-Neutrino (*nt*)

(i) *nt* and $\bar{n}_t$ are produced in the tau decays.

(ii) Experimental upper limit on their mass is 18.2 MeV.

8.5.2 Hadrons

The hadron family contains baryons and mesons:

(i) Baryons are particles having mass in the 938 MeV to few GeV regions and spin half-integral multiple of □. Since they follow Fermi–Dirac statistics, these are known as the fermions.

(ii) Mesons are particles with spin angular momentum either zero or integral multiple of □. They follow Bose–Einstein statistics and are known as the bosons. Some of the properties of the mesons are given in the following paragraph.

The hadrons appear in certain grouping, which are designated by their spin (*J*) and parity (*P*). That means, all the particles in one grouping carry

the same spin and same parity. The list of the hadrons started growing with H. Yukawa's prediction of three pions (p^+, p^0, p^-) in 1934 to explain the nuclear force between protons and neutrons. These were called *mesons* (meaning *middle: Greek*) as their masses lie between that of the electron and the proton. Pions and many such particles were discovered in the pioneering works on cosmic rays, which were known to leave traces in a *cloud chamber*. With the construction of high-energy particle accelerators, hundreds of new and unstable particles, carrying even high spins (up to 6 □ for mesons, and

13/2 □ for baryons), were created in the middle of the 20th century. Collectively these are known as *hadrons* (meaning *strong: Greek*). Most of the observed hadrons having spin higher than 1/2 are found to decay within around 10^{-23} s, and are called *resonances*, whereas the lowest spin states decay through weak interaction or electromagnetic interaction (lifetimes $\sim 10^{-16}$ s). Only proton so far has been found to be stable.

8.5.3 Mesons

These are the particles, whose masses lie between ~130 MeV to few GeV. Mesons with spin zero to 6□ have been observed. So all of them are bosons. Most of them decay through strong interactions, and are called *meson resonances*. Lightest of them, having spin zero and negative parity ($J^P = 0^-$), *p*-mesons (or pions), *K*-mesons (or kaons) and *h*-mesons belong to this group. Obviously they have zero magnetic moments. In Table 8.6, mass, lifetime and dominant decay modes of $J^P = 0^-$ mesons are given. We now discuss the some more properties of these mesons.

TABLE 8.6 Masses, lifetime and dominant decay modes of mesons ($J^P = 0^-$)

Particle	Antiparticle	Mass (MeV)	Lifetime (s)	Dominant decay modes
p^-	p^+	139.6	2.6×10^{-8}	$p^- \rightarrow m^- +$
p^0	p^0	135.0	0.84×10^{-16}	$p^0 \rightarrow 2g$
K^+	K^-	493.7	1.24×10^{-8}	$K^+ \rightarrow m^+ nm/2p$
K^0		497.6	$= 0.90 \times 10^{-10}$	K^0 □ $2p/3p/pen_e$
			$= 5.1 \times 10^{-8}$	

h	h	547.5	$0.5 \square 10^{-18}$	$h^0 \square 3p/2g$

p-Mesons (Pions)

In 1935, H. Yukawa proposed that the strong force between nucleons in a nucleus must be mediated by particles, called *pions* (p), of around 140 MeV mass. Pions exist as positively (p^+), negatively (p^-) and neutral (p^0) states. In 1947, a University of Bristol group was the first to provide the evidence of charged pions by observing their weak decays in the nuclear emulsion photographs. It was also found that the muons arising in these decays possessed a unique energy, thus indicating that the pions decayed into two particles as

$$p^+ \square m^+ + nm \square e^+ + n_e + \bar{n}_m + nm$$

$$p^- \square m^- + \bar{n}_m \square e^- + \bar{n}_e + nm + \bar{n}_m$$

Evidence for the neutral pion was obtained in 1950 through the appearance of electron–positron pairs in the nuclear emulsions, since p^0 decays electromagnetically as

$$p^0 \square g + g$$

and the photons are materialized as electron–positron pairs.

(i) Pions exist as positively (p^+), negatively (p^-) and neutral (p^0) states.
(ii) Pions have also been produced by the following reactions:

$$p + p \square p^+ + d$$

$$p + n \square p^- + p + p$$

$$g + p \square p^0 + p$$

(iii) Mass of $p\square$-meson is 139.9 MeV and that of p^0 is 135.0 MeV.
(iv) Charged pions have several possible weak decay modes, which are:

$$p^+ \square m^+ + nm$$

$$p^- \square m^- + \bar{n}_m$$

$$p\square \square p^0 + e\square + n_e$$

$$p^- \square e^- + \bar{n}_e$$

The first two are the dominant decay modes. p^0 decays through several channels electromagnetically, however, its dominant modes are:

$$p^0 \rightarrow g + g$$
$$p^0 \rightarrow e^+ + e^- + g$$

(v) Lifetime of charged pions is 2.60×10^{-8} s, and that of neutral pion is 0.84×10^{-16} s.

K-Mesons (Kaons)

(i) Kaons exist as positively (K^+), negatively (K^-) charged and their respective neutral partners (K^0 and $\bar{K}^0$). The first evidence of neutral and charged kaons was obtained in 1947 in the cosmic radiation studies with cloud chamber and nuclear emulsion by observing their decays emitting two charged pions ($p^+ + p^-$) or three charged pions ($p^+ + p^+ + p^?$).

(ii) Kaons having mass around 495 MeV are produced in the following reactions:

$$p^- + p \rightarrow K^0 + \Lambda^0$$
$$p^- + p \rightarrow n + K^+ + K^-$$
$$p^+ + p \rightarrow K^+ + \bar{K}^0 + p$$

(iii) Charged kaons decay dominantly through the following leptonic modes:

$$K^+ \rightarrow m^+ + \nu m$$

the first evidence for which was obtained by Paris Group using double cloud chambers, or nonleptonically as

$$K^+ \rightarrow p^+ + p^0$$

(iv) Lifetime of $K^{\pm} = 1.24 \times 10^{-8}$ s.

h-Meson

(i) *h*-meson occurs only in uncharged state.

(ii) *h*-meson can be produced by the following reactions:

$$p^+ + {}^{2}_{1}\mathrm{H} \rightarrow p + p + h$$

$$K^- + p \rightarrow \Lambda + h$$

(iii) h-meson decays through the following modes:

$$h \rightarrow g + g$$
$$h \rightarrow p^0 + p^0 + p^0$$
$$h \rightarrow p^+ + p^- + p^0$$
$$h \rightarrow p^+ + p^- + g$$

(iv) Decay width Γ of h is 1.30 keV, giving its lifetime $t = 0.5 \times 10^{-18}$ s, since

$$t = \frac{\hbar}{\Gamma}$$

8.5.4 Baryons

Following the discovery of several mesons in cosmic rays and laboratory experiments, considerable research was carried out on the heavy unstable particles during the early 1950s, which revealed a rich variety of particles which were classified according to their masses, lifetimes and decays.

In Table 8.7, mass, magnetic moment and lifetime of $J^P = \frac{1^+}{2}$ baryons are given. The spin 1/2 baryons are classified into two categories:

1. Nucleons
2. Hyperons

TABLE 8.7 Mass and lifetime of particles and antiparticles of spin 1/2 baryons

Baryon	Antibaryon	Mass (MeV)	Baryon magnetic moment (nuclear magneton)	Lifetime (s)
p		938.3	2.79	Stable
n		939.6	–1.91	885.7
Λ		1115.7	–0.61	2.6×10^{-10}
Σ^+		1189.4	+2.46	0.80×10^{-10}
Σ^0		1192.6	–	7.4×10^{-20}
Σ^-		1197.4	–1.16	1.48×10^{-10}

Ξ^0	1314.8	–1.25	2.90×10^{-10}
Ξ^-	1321.3	–0.65	1.64×10^{-10}

1. **Nucleons (*p*, *n*):** Protons, neutrons and their antiparticles belong to this category. The proton (meaning *first: Greek*) owes its name to the fact that the simplest nucleus, that of the first element—hydrogen, is just the proton itself. Both these particles have spin 1/2 and hence they are fermions. Under the influence of nuclear force, both proton and neutron are stable inside the stable nuclei. However, a free neutron is unstable and decays as under:

$$n \rightarrow p + e^- + \bar{n}$$

with a half-life of about 886 s, i.e. 15 minutes. However, it acts as stable particle inside the nuclei. Unlike free neutron, a free proton is considered to be a stable particle. Proton and neutron have antibaryon partners called *antiproton* and *antineutron* respectively.

2. **Hyperons:** The term hyperons refer to those unstable particles which carry additional quantum numbers like strangeness to distinguish them from the nucleons. The well-known hyperons are: (lambda), (sigma), (xi) carrying spin ½ and (omega) hyperons, which has spin 3/2.

(i) *-Hyperon*

It was first detected in the cosmic rays in 1947. Later on it was also produced in laboratory by shooting 1.5 GeV p^--mesons in a cloud chamber containing hydrogen gas. In this reaction was produced along with another particle K^0

$$p^- + p \rightarrow + K^0$$

The charge of is zero, and its antiparticle is $\bar{\Lambda}$. Mass of -hyperon was found to be 1115.7 MeV and its half-life is 2.63×10^{-10} s. Main decay modes of -hyperon are:

$\rightarrow p + p^-$

$\rightarrow n + p^0$

Like beta decay, can also emit a lepton pair or even a photon as

$\square \square p + e^{-} + \bar{n}_e$

$\square \square n + g$

though with a less probability. The last decay involves both weak and electromagnetic interactions.

(ii) $\square$-*Hyperons*

There are three $\square$-hyperons $\square^{+}$, $\square^{0}$ and $\square^{-}$; their antiparticles are $\bar{\Sigma}^{+}, \bar{\Sigma}^{0}$ and $\bar{\Sigma}^{-}$. The $\square^{+}$-hyperon was first identified in a cloud chamber in 1953. Next year, the negative counterpart, $\square^{-}$, was observed in a diffusion chamber exposed to negative pion beam of the Brookhaven cosmotron. Lifetime of $\square^{+}$ is $0.80 \square 10^{-10}$ s and mass is 1189.4 MeV. The main decay modes of $\square^{+}$ are the following weak decays:

$$\square^{+} \square \square p + p^{0}$$

$$\square^{+} \square n + p^{+}$$

However, it can also emit a lepton pair

$$\square^{+} \square \square + e^{+} + n_e$$

and even photon in the following decay,

$$\square^{+} \square p + g$$

arising through a mixture of weak and electromagnetic interactions.

Lifetime of $\square^{-}$ is $1.48 \square 10^{-10}$ s and mass is 1197.4 MeV. The main decay mode of $\square^{-}$ is

$$\square^{-} \square n + p^{-}$$

and with less probability it decays to

$\square^{-} \square n + e^{-} + \bar{n}_e$

The neutral member of this family, $\square^{0}$, was observed later in accelerator experiments. Mass of $\square^{0}$ is 1192.6 MeV, its main decay mode is

$$\square^{0} \square \square + g$$

with half-life of $7.4 \square 10^{-20}$ s.

(iii) $\square$-*Hyperons*

□-hyperon also known as cascade hyperon exists in two forms □□$^-$ and □□^{0}; their corresponding antiparticles are $\bar{\Xi}^+$ and $\bar{\Xi}^0$.

□□$^-$ has half-life 1.64 □ 10^{-10} s and mass 1321.3 MeV. It decays dominantly as

$$\square^- \square \square + p^-$$

However, it can also decay through the following decay modes:

$$\square^- \square \square + e^- + \bar{n}_e$$

$$\square^- \square \square + m^- + \bar{n}_m$$

$$\square^- \square \square^- + g$$

It is to be noted that no positively charged counterpart of □$^-$ has ever been observed. The □$^-$ hyperon was postulated to be a charge doublet by Gell-Mann with neutral hyperon □^{0} as its partner. During the period 1954–56, the emphasis shifted from the studies of cosmic rays to experiments on the particle accelerators, in which several above mentioned (e.g. *K*, □, □ and □) and new hadrons were observed. In such experiments,□□^{0} has half-life 2.90 □ 10^{-10} s and mass 1314.8 MeV. Its dominant decay is

$$\square^0 \square \square + p^0$$

though it also decays through the following modes:

$$\square^0 \square \square^+ + m^- + \bar{n}_m$$

$$\square^0 \square \square^+ + e^- + \bar{n}_e$$

$$\square^0 \square \square + g$$

$$\square^0 \square \square^0 + g$$

Here also photon emitting decays occur through combined weak and electromagnetic forces.

(iv) □-*Hyperon*

□$^-$ was discovered in the following reaction:

$$K^- + p \square \square^- + K^+ + K^0$$

It has negative charge, and its main decay modes are:

$$\square^- \square \square^0 + p^-$$
$$\square^- \square \square^- + p^0$$
$$\square^- \square \square + K^-$$

Its half-life is 0.8 $\square$ 0^{-10} s and mass is 1672.4 MeV. Its magnetic moment is –2.02 nuclear magneton.

Baryon Resonance States

A large number of baryon states carrying spin $1/2\hbar$ to $11/2\hbar$ have been observed. Out of these baryons, eight $J^P = 1/2^+$ and ten $J^P = 3/2^+$ are called *low-lying baryons*. Like mesons, most of the baryons with spin $>1/2\hbar$ decay quickly through strong interactions, living around 10^{-23} s, roughly the time taken by light to travel across a proton. These are called *baryon resonances*. Out of the ten $J^P = 3/2^+$ baryons, only $\square^-$ being stable against the strong interactions, decays through the weak interactions.

8.6 QUANTUM NUMBERS

In physics a conservation law states that a particular property of an isolated physical system does not change as the system evolves in time. In classical physics, we know that a process can take place if it is allowed kinematically. We are familiar with some of such properties like energy, momentum, angular momentum, and charge, which are conserved in a classical reaction. These properties also remain conserved in a quantum process or reaction. Besides these, there are certain new properties called *quantum numbers*, which are relevant to microscopic physics only. We do not know all the quantum numbers at the start, their presence is in fact deduced from experimental observations of various decays and reactions involving elementary particles. Therefore, before discussing the conservation laws in particle physics, we discuss those quantum numbers, which are relevant to nuclear and particle physics.

8.6.1 Spin (*S*)

All the elementary particles can be classified as either bosons or fermions depending upon whether their spins are integral multiple of □ or half-odd integral multiple of □ respectively. All quanta of the fundamental interactions and mesons are bosons, whereas the leptons, baryons and quarks are fermions. There exist $2S + 1$ components of the spin along third axis, which are denoted by S_3. Thus, for a spin ½□ particle, say electron, $2S + 1 = 2$, so the possible values of $S_3 = \frac{1}{2}$ (for spin up) and $S_3 = -\frac{1}{2}$ (for spin down).

For composite states, spin adds vectorially to the orbital angular momentum to give rise to total angular momentum,

$$\vec{J} = \vec{L} + \vec{S}$$

that means J can have discrete values starting from $|L - S|$ to $(L + S)$ differing by unity. For example, $L = 2$ and $S = ½$, add to give

$J = 3/2, 5/2$

whereas the third component will have $(2J + 1)$ values.

For example, if $J = \frac{5}{2}$, the values of the third component will be

$$\frac{-5}{2}, \frac{-3}{2}, \frac{-1}{2}, \frac{1}{2}, \frac{3}{2}, \frac{5}{2}$$

8.6.2 Charge (*Q*)

All the elementary particles carry either positive, negative or zero electric charge. Measuring in units of proton charge, they carry +1, –1 or zero (rarely +2) value of the electric charge. Charges of the various elementary particles have already been given in respective tables.

8.6.3 Parity (*P*)

Parity causes reflection of the wave function, i.e. changes x co-ordinate to $-x$, y to $-y$ and z to $-z$, or

$P\psi(\vec{r}) = \psi(-\vec{r})$

If operator P is applied second time, the original wave function is obtained, or

$$P^2\psi(\vec{r}) = P\psi(-\vec{r}) = \psi(\vec{r})$$

Therefore,

$P^2 = 1$ or $P = \pm 1$

If $P = +1$, then wave function is said to have even parity. And if $P = -1$, then wave function is said to have odd parity. Thus, depending upon the behaviour of particle wave functions,

$\psi(-x, -y, -z) = \pm\psi(x, y, z)$

they carry either positive or negative parity quantum number. Mesons (bosons) can carry positive or negative parity. Baryons are assigned positive parity, whereas their antiparticles (antibaryons) carry negative parity. Similarly, antileptons carry opposite parity to leptons. Photon and gluons have negative parity. Since weak interactions can violate parity, no fixed parity can be assigned to *W*- and *Z*-bosons.

Parity is a multiplicative quantum number, i.e. parity of a composite state is given by product of parities of its constitutents.

8.6.4 Lepton Number (*L*)

All the leptons are assigned a lepton number +1 and their respective antiparticles are assigned a lepton number –1. Therefore, lepton number of e^-, m^-, t^-, n_e, *nm* and *nt* is +1, and for their antiparticles, it is –1. In fact, there are three lepton numbers: electron-type, muon-type and tau- type. Thus, the electron and its neutrino n_e have electron-lepton number $L_e = 1$ and positron and antineutrino $\bar{n}_e$ have $L_e = -1$. All other particles have $L_e = 0$. Muon (m^-) and its neutrino *nm* have muon-lepton number $Lm = 1$ and m^+ and antineutrino $\bar{n}_m$ have $Lm = -1$. All other particles have $Lm = 0$, and similarly for the *t*-lepton and its neutrino. Leptons can thus be split into three families, namely (e^-, n_e), (m^-, *nm*) and (t^-, *nt*), each having its own family lepton number. For all other particles, like hadrons and quanta of forces, lepton number is taken to be zero.

8.6.5 Baryon Number (*B*)

All baryons are assigned a baryon number +1 and for the corresponding antiparticles, the baryon number is –1. Thus, baryon number of *p*, *n*, Λ^0,

□$^{+}$, □$^{-}$, □^{0}, □$^{-}$, □^{0} and □$^{-}$ is +1, and their corresponding antiparticles have baryon number as –1. For all other matter particles, like mesons, leptons, and quanta of forces, like photon, gluons, *W* and *Z* bosons, the baryon number is assigned to be zero.

8.6.6 Isospin (*I*)

Besides having same spin, proton and neutron have many similarities. Particularly, it is well known that nuclear forces (strong forces) are charge-independent. That means strong binding force between *n–p, p–n*, and *n–n* is essentially the same. The small difference between them is attributed to the electromagnetic interactions, which arise due to difference in charges of proton and neutron. These facts suggested the existence of new kind of nuclear spin called *isospin.* Stating differently, in the world of strong interactions only, we can think of proton and neutron as the two orthogonal states of the same particle called *nucleon*[9].

9. This is very similar to the ordinary spin-up and spin-down states of an electron, which degenerate till some magnetic field is applied.

The isospin is a quantity introduced to describe groups of particles (hadrons) which have nearly same mass. A general isospin state is described by two quantum numbers, I and I_3, where I_3 can have values ranging from $-I$ to I separated by unity. Projections of isospin along the third axis, I_3, appear as different charge states of the corresponding particle. For a group of two particles or doublet (like proton and neutron), $I = \frac{1}{2}$, and for triplet (like □$^{+}$, □$^{-}$ and □^{0}) $I = 1$. Similar to spin, there are $2I + 1$ components of the isospin along third axis, which are distinguished by I_3. Therefore, for a doublet (*p*□*–n*) $2I + 1 = 2$, so the possible values of $I_3 = \frac{1}{2}$ (for proton) and $I_3 = -\frac{1}{2}$ (for neutron).

Among mesons, K^{+} and K^{0} form isospin doublet. Similarly, for a triplet (□$^{+}$, □^{0}, □$^{-}$), isospin is described by isospin vector, i.e. $I = 1$. Therefore, I_3

= +1 corresponds to □$^{+}$, I_3 = 0 corresponds to □^{0} and I_3 = –1 corresponds to □$^{-}$. Among mesons, pions (p^{+}, p^{0}, p^{-}) form isospin vector.

One may observe an interesting relation among the electric charge, third component of isospin and baryon number

$$Q = I_3 + \frac{B}{2}$$

which gives the charge (Q) present on each member of the isospin multiplet. For example, baryon number of p–n multiplet is +1 and I_3 for proton is $+\frac{1}{2}$. Therefore, $Q = \frac{1}{2} + \frac{1}{2} = +1,$ indicating that +1 is the charge present on proton. For neutron I_3 is $-\frac{1}{2}.$ Therefore, $Q = -\frac{1}{2} + \frac{1}{2} = 0,$ indicating that no charge is present on neutron.

8.6.7 Strangeness (*S*)

Soon after the discovery of the hyperons, it was realized that these particles were behaving in a strange manner. They are produced through the strong interactions, and have large production cross-sections. However, they have lifetimes characteristics of the weak interactions. Their decays are always by way of weak interaction (or in few cases, weak electromagnetic interactions). Another strange fact about these particles is that they are always produced in pairs, e.g. *K*-meson in association with □ or □-hyperon. This effect was called associated production by A. Pais. For instance, *K*-meson and □-baryon were discovered in a pion-nucleon reaction,

$$p^{-} + p \rightarrow \square(\square\, p + p^{-}) + K^{0}(\square\, p^{+} + p^{-})$$

These are called *strange* particles as they are produced in pairs through strong interaction (lifetimes ~10^{-23} s), but decay through weak interactions (lifetimes ~10^{-8} s). This situation led to use of the expression "Strange particles" to describe *K*-mesons and hyperons, as all this was certainly puzzling. This finally led to suggest that a new quantum number might be associated with these new particles, which was called *strangeness* (*S*) by Gell–Mann and Nishijima in 1953 to account for the strange behaviour of

K-mesons and hyperons. This quantum number was conserved in the strong interactions, but not in weak decays. A nonzero strangeness, is assigned to them in contrast to the nucleons, pions and photon, which have zero strangeness, $S = 0$.

For the remaining particles, strangeness is assigned using the Gell–Mann–Nishijima relation given below.

8.6.8 Gell–Mann–Nishijima Scheme

Gell–Mann–Nishijima observed that the electric charge (in units of e) of a hadron can be related to its isospin, baryon number and strangeness as follows:

$$Q = I_3 + \frac{B+S}{2} \tag{8.1}$$

For instance, for proton, we have $Q = 1$, $I_3 = ½$, $B = 1$ and $S = 0$, which obviously satisfy this relation. Similarly, quantum numbers of positively charged pion, i.e. $Q = 1$, $I_3 = 1$, $B = 0$ and

$S = 0$, satisfy it. Using the relation, strangeness of various particles can be assigned. For example, for □-hyperon, using $Q = 0$, $I_3 = 0$ and $B = +1$ Gell–

Mann–Nishijima relation gives strangeness $S = -1$. Similarly, for □□¯-hyperon, carrying $Q = -1$, $I_3 = 0$ and $B = +1$, the relation yields

$$S = 2(Q - I_3) - B = 2(-1 - 0) - 1 = -3$$

In Table 8.8, we have given masses and various quantum numbers like charge, baryon

number, isospin and strangeness of the mesons. Similarly, properties of the baryons are given in Tables, 8.9(a) and 8.9(b).

TABLE 8.8 Quantum numbers of mesons

Particle	Mass	Spin	Charge	Isospin	Isospin	Baryon	Strangeness	Hypercharge
	(MeV)	(□)	(Q)	(I)	(I_3)	number (B)	(S)	(Y)
K^+	493.7	0	+1	1/2	+1/2	0	+1	1
K^0	497.6	0	0	1/2	–1/2	0	+1	1
p^+	139.6	0	+1	1	+1	0	0	0
p^0	135.0	0	0	1	0	0	0	0

p^-	139.6	0	–1	1	–1	0	0	0
	497.6	0	0	1/2	+1/2	0	–1	–1
K^-	493.7	0	–1	1/2	–1/2	0	–1	–1
h	547.5	0	0	0	0	0	0	0

TABLE 8.9(a) Quantum numbers of baryons $J^P = \frac{1^+}{2}$

Particle	Mass (MeV)	Spin (□)	Charge (Q)	Isospin (I)	Isospin (I_3)	Baryon number (B)	Strangeness (S)	Hypercharge (Y)
p	938.6	1/2	1	1/2	+1/2	1	0	1
n	939.6	1/2	0	1/2	–1/2	1	0	1
□	1115.6	1/2	0	0	0	1	–1	0
□$^+$	1189.4	1/2	1	1	+1	1	–1	0
□^{0}	1192.6	1/2	0	1	0	1	–1	0
□$^-$	1197.4	1/2	–1	1	–1	1	–1	0
□^{0}	1314.7	1/2	0	1/2	+1/2	1	–2	–1
□$^-$	1321.2	1/2	–1	1/2	–1/2	1	–2	–1

TABLE 8.9(b) Quantum numbers of $J^P = \frac{3^+}{2}$ baryons

Particle	Mass (MeV)	Spin (□)	Charge (Q)	Isospin (I)	Isospin (I_3)	Baryon number (B)	Strangeness (S)	Hypercharge (Y)
□$^{++}$	1232	3/2	2	3/2	3/2	1	0	1
□$^+$	1232	3/2	1	3/2	+1/2	1	0	1
□^{0}	1232	3/2	0	3/2	–1/2	1	0	1
□$^-$	1232	3/2	–1	3/2	–3/2	1	0	1
□$^{*+}$	1382.8	3/2	1	1	+1	1	–1	0
□^{*0}	1383.7	3/2	0	1	0	1	–1	0
□□$^-$	1387.2	3/2	–1	1	–1	1	–1	0
□^{*0}	1531.8	3/2	0	1/2	+1/2	1	–2	–1
□$^{*-}$	1535.0	3/2	–1	1/2	–1/2	1	–2	–1
□$^-$	1672.5	3/2	–1	0	0	1	–3	–2

Masses of □-baryons lie between 1231 MeV and 1233 MeV. Due to their large decay width of 118 MeV (equivalently very small lifetimes □10^{-23} s) of □ baryons, their masses are not known with greater precision than given here.

8.6.9 Hypercharge (*Y*)

Gell–Mann–Nishijima relation can also be written as

$$Q = I_3 + \frac{Y}{2} \qquad (8.2)$$

where

$$Y = B + S$$

is called *hypercharge*. All the $2I + 1$ members of a given isospin multiplet (also called *isomultiplet*) have the same hypercharge Y. For example, for the triplet p^+, p^0 and p^-, the hypercharge is zero. For K-mesons, $I = 1/2$, there are two multiplets, one consisting of the particles K^+ and K^0, and the other of the corresponding antiparticles $\bar{K}^-$ and $\bar{K}^0$, the hypercharge of the former pair is +1 and that of the latter pair is –1. Leptons, photon, gluons do not carry hypercharge, i.e. their hypercharge is zero. It is interesting to note that all the mesons and baryons shown in the tables given above carry similar patterns for electric charge and hypercharge. For these hadrons, one may also note that Y is also equal to twice the average charge of the members of the isospin multiplet.

Charm, Bottom and Top Quantum Numbers

Subsequent discovery of new particles, like D, D_S, B and $\square_c$ hadrons, introduced new quantum numbers, called *charm* (C), bottom (b), and top (t)[10]. Including these quantum numbers, definition of the hypercharge has been extended to

10. Some people like to call beauty and truth in place of bottom and top quantum numbers.

$$Y = B + S + C + b + t,$$

thus extending the Gell–Mann–Nishijima formula to

$$Q = I_3 + \frac{B + S + C + b + t}{2} \qquad (8.3)$$

keeping the form (8.2) unaffected. This permits us to give an alternative definition of the hypercharge, as twice the difference between the electric (Q) charge and isospin component (I_3) of the particle, i.e.

$$Y = 2(Q - I_3) \qquad (8.4)$$

A particle and its antiparticle have opposite value of both Q and I_3, hence the hypercharge will be numerically equal in magnitude but opposite in sign, or zero.

8.7 CONSERVATION LAWS

All the interactions are controlled by certain conservation laws. In Section 4.4, conservation laws for nuclear reactions have been discussed. However, there exist a few conservation laws, which are valid for some interactions and not for other interactions. For example, parity, isospin, and other flavour quantum numbers are not conserved in the weak interactions. There happens to exist a rule: the stronger an interaction the more symmetrical it is.

The familiar conservation laws of classical physics, i.e.

- Conservation of energy
- Conservation of charge
- Conservation of linear momentum
- Conservation of angular momentum

also hold good for particle physics, provided we have made relativistic formula for to mass, momentum, etc.

8.7.1 Conservation of Charge

The electric charge is exactly conserved in particle decays and their reactions. It is due to this conservation law that electron, the lightest

charged particle cannot decay, for instance, via

$$e^- \rightarrow n_e + g$$

Note that final state has null charge in contrast to the initial state particle. That is why, electron is treated as stable particle, which is supported by the experimental lower limit on lifetime of electron for this decay

$$t_e > 4.6 \times 10^{26} \text{ years}$$

Some other conservation laws, which are relevant to quantum physics in general and particle physics in particular are discussed below.

8.7.2 Conservation of Lepton Number

This law states that in any particle reaction/decay, the total lepton number is always conserved. Consider the decay of m^+-meson,

$$\begin{array}{cccccc} \mu^+ & \rightarrow & e^+ & + & \nu_e & + \bar{\nu}_\mu \\ -1 & & -1 & & 1 & -1 \end{array} \qquad \text{(lepton number)}$$

In this decay, lepton number before and after the decay is –1 and hence lepton number is conserved. One may verify the electron-lepton number and muon-lepton number are also separately conserved in this decay process.

Let us now consider the decay of neutron

$$n \rightarrow p + e^- + \bar{n}_e$$

is allowed, since lepton number before the decay is zero and it is also zero (0 + 1 – 1 = 0) after the decay. However, the following decay

$$n \rightarrow p + e^- + n_e$$

is not allowed, since total lepton number of products is 0 + 1 + 1 = 2, whereas the lepton number before the neutron decay is zero. So conservation of lepton number does not permit this decay to occur.

8.7.3 Conservation of Baryon Number

This law states that in any particle reaction/decay, the total number of baryons is always conserved. Consider the decay

$$\square \square p + p^{-}$$

baryon number of $\square$ is +1 and that of p is also +1, while that for p^{-}, it is zero. Therefore, the baryon number before and after decay is +1 and hence this decay is allowed. However, in the following decay

$\square \square \bar{p} + p^{+}$

the baryon number before the decay is +1 while after the decay it is –1, hence this decay is not allowed. Similarly, for the reaction

$$p^{+} + n \square K^{0} + K^{+}$$

baryon number of the reactants is +1 while that of the products it is zero. Hence, this reaction is not allowed. No decay or process has yet been observed in which baryon number is not conserved. The stability of proton, which is the lightest baryon, provides a critical test of the exact conservation of baryon number.

Conservation laws mentioned above are valid in all the four interactions, i.e. strong, weak, electromagnetic and gravitational interactions. No exception to these conservation laws has been found in any particle reactions. However, there are certain conservation laws, which are valid for strong interactions only. Electromagnetic interactions can violate isospin conservation. Several quantum numbers, like isospin, strangeness, charm, parity, C-parity, may not be conserved for weak interactions.

8.7.4 Conservation of Isospin (*I*)

According to law of conservation of isospin, the total isospin for particles subjected to strong interactions is always same before and after the particle reaction/decay. Remember that like angular momentum addition, isospin adds vectorially

$$\vec{I} = \vec{I}_1 + \vec{I}_2$$

that means I can have discrete values starting from $| I_1 - I_2 |$ to $(I_1 + I_2)$ differing by unity. For example, isospin 1/2 when added to isospin 1, would give 1/2 and 3/2 as total isospin. Consider the following decay:

$$\Delta^{+} \rightarrow p + \pi^{0}$$

3/2 1/2 1 (total isospin, I)

in which total isospin of the final state (1/2 + 1 = 3/2) match with that of the parent particles, so this decay is allowed to occur through the strong interactions. Now consider the decay of p^0

$p^0 \not\rightarrow g + g$

Before the decay isospin is 1 and after the decay, it is zero. Hence, this decay is not allowed by strong interaction. However, this decay is possible via the electromagnetic interactions, as total isospin may not be conserved in these interactions.

8.7.5 Conservation of Component of Isospin (I_3)

According to this conservation law, for strong and electromagnetic interactions I_3, the third component of isospin before and after particle reaction/decay always remains the same. Consider the following reaction:

$$p + p \rightarrow \Lambda^0 + K^0 + p + \pi^+$$
$$1/2 \quad 1/2 \quad 0 \quad -1/2 \quad 1/2 \quad 1 \text{ (third component of isospin, } I_3)$$

Total I_3 component before the reaction is 1/2 + 1/2 = 1, After the reaction total I_3 component is 0 – ½ + ½ + 1 = 1. The component I_3 is same before and after the reaction, and hence this reaction is possible. Similarly, for the following decay:

$$\Delta^+ \rightarrow p + \pi^0$$
$$1/2 \quad 1/2 \quad 0 \quad \text{(third component of isospin, } I_3)$$

However, for the decay

$$\Lambda \rightarrow p + \pi^-$$
$$0 \quad 1/2 \quad -1 \quad \text{(third component of isospin, } I_3)$$

Component I_3 before the decay is zero and after the decay 1/2 – 1 = –1/2, therefore, I_3 is not conserved in this reaction. Hence, this reaction cannot proceed via strong or electromagnetic interactions but can take place through weak interactions.

8.7.6 Conservation of Hypercharge

According to this conservation law, for strong and electromagnetic interactions, the total hypercharge remains the same before and after the particle reaction/decay. Consider the reaction

$$p + p \rightarrow \Sigma^0 + K^0 + p + p^+$$

Hypercharge before the reaction is 1 + 1 = 2 and after the reaction, it is 0 + 1 + 1 + 0 = 2. Hypercharge before and after the reaction is same (= 2), hence this reaction is possible via strong or electromagnetic interactions. However, weak interactions can violate hyperchange conservation like $\Sigma \rightarrow p + p^-$.

8.7.7 Conservation of Parity (*P*)

According to conservation of parity, if a process is subjected to a reflection in mirror (also called inversion), so that the sign of all its Cartesian coordinates are changed

$$\vec{x} \rightarrow \vec{x}' = -\vec{x}$$

the resulting process will be indistinguishable from the original one. In other words, the conservation of parity requires that if an event is possible, its reflection in a mirror represents an equally probable event. It was found that in strong and electromagnetic interactions parity is always conserved. Until 1956, it was thought that parity must always be conserved. However, Lee and Yang in 1956 suggested that parity is not conserved in weak interactions. Examples of such cases are *b*-decay, decay of muons, kaons, etc.

8.7.8 Conservation of Charge Conjugation Parity (*C*)

According to this symmetry principle, a particle is converted into its corresponding antiparticle,

Particle $\rightarrow$ Antiparticle

or

Antiparticle $\rightarrow$ Particle

In other words, a given process or event should be indistinguishable from one involving the corresponding antiparticles. The charge conjugation operator is normally denoted by *C*. Like parity, charge conjugation

symmetry is applicable to strong and electromagnetic interactions only. For weak interactions it fails. A combination of *C*-parity with isospin is called *G*-parity which is conserved only in the strong interactions. Like *P*-parity, *C*-parity is also multiplicative in nature. Combining the two parities, *CP* can also be defined for particles and their processes.

8.7.9 Time Reversal (*T*) Symmetry

According to this symmetry principle, an event on a particle scale should be exactly reversible in time, i.e.

$$t \to t' = -t$$

If a motion picture film was made of an event of say a particle formation or decay, the event would appear to be equally possible regardless of whether the film is run forward or backward.

8.7.10 *CPT* Theorem

According to postulates of advanced quantum mechanics (or quantum field theory), all interactions should be invariant under the combination of *C*, *P* and *T*, i.e. *CPT* operation,

$$\psi_{\text{particle}}(\vec{x}, t) \rightarrow \psi_{\text{antiparticle}}(-\vec{x}, -t)$$

This is also known as *CPT* theorem. Consequences of this theorem have been discussed in Section 8.4.3.

8.7.11 Permutation Symmetry and Spin Statistics Theorem

The elementary particles are also called identical particles, i.e. no observations can distinguish particles of the same kind. This is typical quantum feature which has no classical analogue. The symmetry under permutation (or exchange of identical particles) operation in the composite state allows particular combinations only. This is intimately connected with structure of their wave functions. As per the spin-statistics theorem, quantum mechanical wave function of a system of identical bosons must be symmetric under the exchange of any pair of the particles, i.e.

$$y_B(x_1, x_2, x_3, \ldots) = y_B(x_2, x_1, x_3, \ldots)$$

where x_i denotes collectively space–time co-ordinates and internal quantum numbers. Thus, bosons follow Bose–Einstein statistics. Similarly, the

quantum mechanical wave function of a system of identical fermions must be antisymmetric under the exchange of any pair of the particles, i.e.

$$y_F(x_1, x_2, x_3,\ldots) = -y_F(x_2, x_1, x_3,\ldots)$$

and follow Fermi–Dirac statistics.
For instance, for two particle states, we have the following permitted states:

- Symmetric state for the bosons

$$\psi_S = \frac{1}{\sqrt{2}}[\psi(1, 2) + \psi(2, 1)]$$

- Antisymmetric state for the fermions

$$\psi_A = \frac{1}{\sqrt{2}}[\psi(1, 2) - \psi(2, 1)]$$

8.8 WEAK DECAYS OF STRANGE PARTICLES

In general, leptons and hadrons given in Section 8.5 decay through weak interaction. These are classified in the following three varieties.

Leptonic Decays

In such decay modes, only leptons are involved. For example,

$$\text{Muon decays: } \mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu$$
$$\text{Tau-lepton decays: } \tau^- \rightarrow \mu^- + \bar{\nu}_\mu + \nu_\tau$$

Semileptonic Decays

In these decay modes, leptons as well as hadrons are involved. For example,

$$\text{Pion decays: } \pi^- \rightarrow \mu^- + \bar{\nu}_\mu$$
$$\text{Neutron-beta decays: } n \rightarrow p + e^- + \bar{\nu}_e$$

Nonleptonic Decays

These decays are also called hadronic decays, as only hadrons are involved in such decays. For example,

K-decays: $K^0 \rightarrow \pi^+ + \pi^-$

Lambda decays: $\Lambda \rightarrow p + \pi^-$

Sometimes a photon is also emitted in hadronic weak decays. Such decays take place by combined effects of weak and electromagnetic interactions. For example,

□ □ $n + g$

8.9 PARTICLE SYMMETRIES

The concept of symmetry has acquired great and fundamental significance in classical as well as in modern physics, more particularly in particle physics. The term symmetry means that a system, state or quantity remains unchanged (or invariant) as a result of a particular transformation, such as a change in space coordinates, or change in time, etc. Symmetries also predict degeneracy between different physical states, which can be connected through the corresponding transformations. Every conservation law is related to a particular invariance (or symmetry) principle. For example, conservation of total energy is due to the invariance of a system under shift in time, called *time translation*. Conservation of linear or angular momentum is due to invariance under displacement in space (space translation) or rotation in space respectively. These transformations can be abstract also, i.e. they may have no relation to actual space, time, etc. Examples of these are like conservation of charge, lepton number, baryon number, isospin etc. Besides these continuous symmetries, there also exist discrete symmetries like parity and
C-parity.

In the following, we will describe briefly some aspects of special symmetries in particle physics, which are SU(2) symmetry, SU(3) symmetry, and higher symmetries.

8.9.1 SU(2) Symmetry

Besides having same spin, proton and neutron have many similarities. Particularly, it is well known that nuclear forces (strong forces) are charge-independent. That means strong binding force between *n–p, p–n,* and *n–n* is essentially the same. The small difference between them is attributed to the electromagnetic interactions, which arise due to difference in charges of

proton and neutron. These facts suggested the existence of new kind of nuclear spin, called *isospin*, originally proposed by Heisenberg in 1934. Stating differently, in the world of strong interactions, we can think of proton and neutron as the two orthogonal states of the same particle, called *nucleon*. As has been mentioned before, a general isospin state is described by two quantum numbers, I and I_3, where I_3 can have values ranging from $-I$ to I separated by unity. Thus, for isospin I, there are $2I + 1$ degenerate states available. Isospin conservation is a consequence of the invariance of the strong interaction Hamiltonian under the rotations in isospin space. Isospin symmetry also reflects in the physical states belonging to an isospin multiplet, which have the same mass. Consequently, the nucleon doublet, i.e. proton and neutron, have the same mass due to the strong isospin SU(2) symmetry[11]. SU(2) denotes special (S) unitary (U) group in two complex dimensions, that means it is a group of 2 □ 2 unitary matrices U with unity determinant. A unitary matrix is that whose inverse is given by complex conjugate of its transpose. Such a matrix, can be written as

11. It may be remarked that there also exists weak isospin symmetry needed to develop quantum field theory of weak interactions, but it is beyond the scope of the book.

$$U = \exp\left(i\sum_{j=1}^{3}\sigma_j\theta_j\right)$$

where s_j are the well-known Pauli matrices,

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},\ \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix},\ \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

and q_j denotes the angles of rotation in isospin space. SU(2) allows all integral $(2I + 1)$ multiplets;

$$1, 2, 3, 4, 5,\ldots$$

respectively for $I = 0, 1/2, 1, 3/2, 2,\ldots$ states. The fundamental particle-multiplet of the SU(2) is two-dimensional which is given by two-component column matrices. For example, we can imagine proton and neutron to be described by the following isospin states of the isospin symmetry:

$$p = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, n = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

on which the Pauli matrices can operate.

Strong interactions respect the isospin symmetry leading to conservation of isospin in strong decays or reactions. This is quite similar to the conservation of angular momentum due to the invariance of the Hamiltonian under the rotations in ordinary space (more precisely in spin space). Isospin symmetry is broken by the electromagnetic and weak interactions. However, the electromagnetic interactions conserve the third component of the isospin. Breaking of isospin symmetry permits us to differentiate between a proton and a neutron.

Similarly, strong force treats three pions the same way. It is reflected in the cross-sections being the same for scattering of different pions on proton and neutron. In the isospin space, we can describe three pions as

$$\pi^+ = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \pi^0 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \pi^- = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Looking at Tables 8.8 and 8.9(a, b) giving properties of hadrons, one may notice the states belonging to one isomultiplet are nearly mass degenerate. The small mass difference in the so-called isomultiplets is assigned to the difference in the electric charges of these states.

8.9.2 SU(3) Symmetry

The number of observed hadrons is more than hundred and more of them are yet to be seen. All the observed hadrons are extended objects having size at the Fermi scale. Obviously, all of the hadrons could not be treated as elementary. A certain pattern, similar to the Mendeleev periodic table of atoms, among them was found independently by Gell–Mann and Y. Ne'eman. Consider a system of coordinates in which x-axis and y-axis are used to represent third component of isospin (I_3) and Hypercharge ($Y = B + S$) respectively. If we draw mesons of same spin and parity, say $J^P = 0^-$, they create geometric symmetric octet patterns, called weight diagrams (Figure 8.2). Similar pattern is also created by $J^P = 1/2^+$ baryons, see

Figure 8.3. Gell–Mann called this scheme[12] 'Eightfold Way' as term octet was occurring again and again. In fact, all the observed baryons, known till then, seemed to appear in the grouping of 1 (singlet), 8 (octet) and 10 (decuplet), whereas the mesons fall into groups of 1 and 8. The observed patterns, shown in Figures 8.2 and 8.3, were naturally explained through the SU(3) symmetry, a mathematical generalization of the SU(2) isospin symmetry. SU(3) is the group of 3 □ 3 unitary matrices with unit determinant, and its fundamental multiplet is obviously three-dimensional. In contrast to SU(2), SU(3) permits selective multiplets, say

[12]. M. Gell–Mann called this symmetry 'Eightfold Way', borrowing the name from the Buddhist way to Nirvana.

1, 3, 3*, 6, 6*, 8, 10, 10*, 15,…

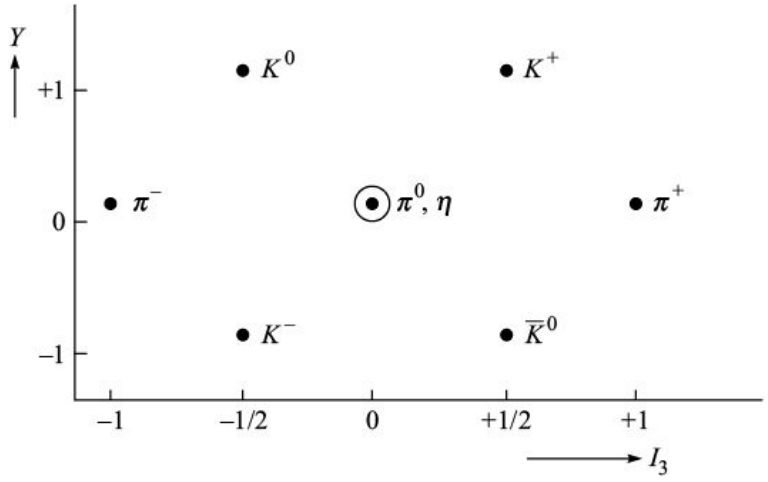


Figure 8.2 Weight diagram of $J^P = 0^-$ mesons.

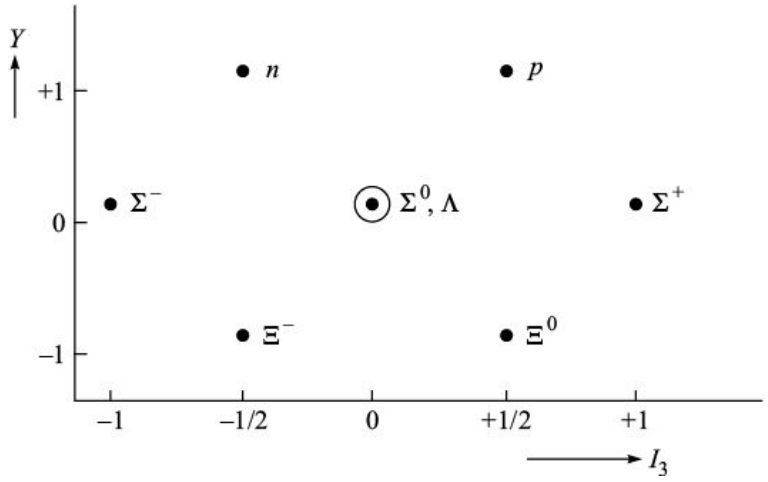


Figure 8.3 Weight diagram of $J^P =$ baryons.

The SU(3) could explain several regularities observed in the hadron world. In fact, the SU(3) symmetry scheme achieved wonderful confirmation with the discovery of □⁻-hyperon, which was predicted by Gell-Mann, as a missing partner of the already observed nine spin 3/2 baryons filling the decuplet. However, this symmetry is not exact and is broken even by the strong interactions, though in a particular way. Assuming that the SU(3) breaking also follows eightfold way, several properties of the hadrons could be explained. For instance, Gell-Mann–Okubo obtained a well-satisfied mass formula:

$2N + 2\square = 3\square + \square$

1129 MeV 1135 MeV

where particle symbol denotes the average mass of the particles in that isospin multiplet.

8.9.3 Higher Symmetries

Later higher symmetries, described by SU(4), SU(5), SU(6) and higher groups, have also been employed to study the hadronic behaviour. For instance, SU(6), which combines the SU(3) symmetry with the SU(2) spin symmetry, could explain reasonably the ratio of nucleon magnetic moments

$m_p/m_n = -3/2$

Experimental value = –1.46

SU(4) and SU(5) are used to describe spectra of hadrons carrying charm and bottom quantum numbers.

8.10 QUARKS

The unitary symmetries were only to act as intermediate step to further explore the structure of matter. With the successes of the SU(3) symmetry, new questions were also raised. For instance, why nature allows only 1, 8, and 10 multiplets for the hadrons, while SU(3) group does possess lower multiplets like 3, 3*, 6 and 6*. This led to the proposal that the hadrons may have constituents which fill these multiplets. Historically, M. Gell–Mann[13]

in 1964 suggested the idea of quarks to explain the existence of a large number of baryons (of half-integral spin) and meson (of integral spin) states. He further suggested that the hadrons are composed of three varieties of quarks, up (*u*), down (*d*) and strange (*s*). The variety of quarks is now formally called *flavour*. Like leptons, all the quarks carry spin ½□. Quarks fill the fundamental multiplet 3 of the SU(3), called *flavour symmetry*. Accordingly, antiquarks were assigned to 3* multiplet:

13. The idea of quarks was introduced in 1962 independently by M. Gell–Mann and G. Zweig. However, Zweig called them *Aces*.

3 = , 3* = (, ,)

With this scheme, he explained the existence of all the multiplets, he then observed baryons (as three-quark states) and mesons (as quark–antiquark states). Thus, regularities in the observed hadron spectrum could be understood through the following group-theoretical decompositions (generalization of concept of addition of angular momenta) of the direct-products into the direct-sum of the multiplets under the SU(3) group:

Mesons (q) □ 3 □□3* = 1 □ 8, (8.5)

Baryons (qqq) □ 3 □ 3 □ 3 = 1 □ 8 □ 8 □ 10. (8.6)

Thus, mesons form octet and singlet, whereas the baryons belong to singlet, octet and decuplet.

Charm, Bottom and Top Quarks

Discovery of *y*(3096 MeV) meson in 1974 required a new quark, named *charm,* as it could not be accommodated within the three-quark scheme. Similarly, observation of *Y* (9460 MeV) meson in 1978 demanded the existence of a fifth quark, named *bottom.* Some of the hadrons, like *D*-and *B*-mesons and , □□□$_c$, baryons containing one of these quarks have been produced in the high-energy experiments. These developments indicated that nature seems to take the basic pattern of the first generation of two quarks and two leptons, and recreate it at higher energies. So the quark-lepton analogy, namely the number of quarks and leptons be the same, suggested the existence of a sixth quark, called *top*, which has been created by colliding high energy

(500 GeV) p and beams at Fermilab in 1994. In all, there are six types of quarks, three with two thirds the charge of a proton and the other three with one thirds the charge of an electron. All six quarks have different masses and other quantum numbers (Table 8.10). Note that mass of the top-quark, which is an elementary particle, is even larger than that of the gold atom.

TABLE 8.10 Properties of quarks

Quark	Name	Mass (GeV)	Charge (e)	Spin (□)	Baryon number (B)	Isospin I	Isospin I_3	Strangeness (S)	Charm (C)	Bottom (beauty) (B)	Top (truth) (t)
d	Down	0.33						0	0	0	0
u	Up	0.33						0	0	0	0
s	Strange	0.51				0	0	–1	0	0	0
c	Charm	1.60				0	0	0	+1	0	0
b	Bottom (beauty)	4.7				0	0	0	0	–1	0
t	Top (truth)	174.2				0	0	0	0	0	+1

Like leptons, the quarks are divided into three generations: (u, d), (c, s), and (t, b). A generation consists of a pair of leptons (one charged and one neutral) and a pair of quarks (carrying charges of 2/3 and –1/3). Stable matter is made of the first generation which contains the electron, the electron neutrino, the u-quark, and the d-quark. The reason for the existence of other generations are not known except that minimum number of generations must be three to allow CP-violation, which seems to play a significant role to explain matter generation in the universe, and to explain small amount of antimatter in it. There are six antiquarks also. They are , , , , and . Their basic properties are shown in Table 8.11.

TABLE 8.11 Properties of antiquarks

Quark	Name	Mass (GeV)	Charge (e)	Spin (□)	Baryon number (B)	Isospin I	Isospin I_3	Strangeness (S)	Charm (C)	Beauty (Truth) (B)	(T)
	Down	0.33						0	0	0	0
	Up	0.33						0	0	0	0
	Strange	0.51				0	0	1	0	0	0
	Charm	1.6				0	0	0	–1	0	0
	Beauty	4.7				0	0	0	0	1	0
	Truth	174.2			–	0	0	0	0	0	–1

8.11 QUALITATIVE DESCRIPTION OF QUARK MODEL

By looking at the quantum numbers of the hadrons and the quarks, one can understand easily their quark composition. For example, positive p meson has one up (u) quark and one antidown ( ) quark. Quark compositions of the octet mesons are given in Table 8.12.

TABLE 8.12 Quark composition of octet mesons

Particle	Quark content	Charge (Q)	Baryon number (B)	Isospin (I_3)	Strangeness (S)
K^+					0 + 1 = 1
K^0					0 + 1 = 1
p^+					0 + 0 = 0
p^0					0 + 0 = 0
p^-					0 + 0 = 0
					–1 + 0 = –1
K^-					–1 + 0 = –1
h					0 + 0 = 0

Similarly, a neutron contains two d-quarks and one u-quark, and proton carrying positive charge is made up of two u-quark and d-quark[14]. It is obvious from the quark combinations to build these nucleons because the u-quark carries electric charge[15] two thirds of the charge on a proton, and d-quark carries one third of the charge on an electron. Obviously, baryon number of quarks is 1/3. For example, charge, baryon number, and strangeness of the proton are given by sum of the respective properties of the constituent quarks

14. Three antiquarks can also combine to form antibaryon states. For instance, two anti-u-quarks and an anti-d-quark would give rise to antiproton.

15. Before the introduction of the quark model, charge of an electron and that of a proton were taken to be the basic units of negative and positive charges, respectively. Now it has been replaced with one third of the proton (or electron) charge.

$$\text{Charge on proton} = \frac{2}{3} + \frac{2}{3} - \frac{1}{3} = +1$$

$$\text{Baryon number} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$$

$$\text{Strangeness} = 0 + 0 + 0 = 0$$

Similarly, quark contents of various baryons and antinucleons are given in Tables 8.13, 8.14 and 8.15. Higher spin states were found to be orbitally

excited states of these quark configurations, somewhat similar to the excited states of hydrogen atom.

TABLE 8.13 Quark composition of octet baryons

Particle	Quark content	Charge (Q)	Baryon number (B)	Isospin (I_3)	Strangeness (S)
p	*uud*	$+\frac{2}{3}+\frac{2}{3}-\frac{1}{3}=+1$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	$+\frac{1}{2}+\frac{1}{2}-1=+\frac{1}{2}$	$0+0+0=0$
n	*udd*	$+\frac{2}{3}-\frac{1}{3}-\frac{1}{3}=0$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	$+\frac{1}{2}-\frac{1}{2}-\frac{1}{2}=-\frac{1}{2}$	

0 + 0 + 0 = 0					
□	*uds*	$+\frac{2}{3}-\frac{1}{3}-\frac{1}{3}=0$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	$\frac{1}{2}-\frac{1}{2}+0=0$	0 + 0 – 1 = –1
$\square^{+}$	*uus*	$+\frac{2}{3}+\frac{2}{3}-\frac{1}{3}=+1$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	$\frac{1}{2}+\frac{1}{2}+0=+1$	0 + 0 – 1 = –1
$\square^{0}$	*uds*	$+\frac{2}{3}-\frac{1}{3}-\frac{1}{3}=0$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	$\frac{1}{2}-\frac{1}{2}+0=0$	0 + 0 – 1 = –1
$\square^{-}$	*dds*	$-\frac{1}{3}-\frac{1}{3}-\frac{1}{3}=-1$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	$-\frac{1}{2}-\frac{1}{2}+0=-1$	0 + 0 – 1 = –1
$\square^{0}$	*uss*	$+\frac{2}{3}-\frac{1}{3}-\frac{1}{3}=0$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	$\frac{1}{2}+0+0=\frac{1}{2}$	0 – 1 – 1 = –2
$\square^{-}$	*d*ss	$-\frac{1}{3}-\frac{1}{3}-\frac{1}{3}=-1$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	$-\frac{1}{2}+0+0=-\frac{1}{2}$	0 – 1 – 1 = –2

TABLE 8.14 Quark composition of antinucleon

Particle	Quark content	Charge (Q)	Baryon number (B)	Isospin (I_3)	Strangeness (S)
		$+\frac{2}{3}+\frac{2}{3}+\frac{1}{3}=+1$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	$\frac{1}{2}+\frac{1}{2}-\frac{1}{2}=\frac{1}{2}$	0 + 0 + 0 = 0
		$-\frac{2}{3}-\frac{1}{3}-\frac{1}{3}=-1$	$-\frac{1}{3}-\frac{1}{3}-\frac{1}{3}=-1$	$-\frac{1}{2}-\frac{1}{2}+\frac{1}{2}=-\frac{1}{2}$	0 + 0 + 0 = 0

The quark model shattered two basic concepts about fundamental particles. Firstly, a proton and a neutron were excluded from the family of fundamental particles and secondly, the charge of an electron did not remain the minimum possible value for the electric charge of any particle. Quarks were considered to be the key components in the quark model. Phenomenological analysis of hadron properties, like masses, magnetic moments and decays, using the quark model has been very useful. For instance, $n \rightarrow p + e^{-} + \bar{\nu}_e$, at quark level, involves the conversion of *d*-type of quarks through $d \rightarrow u + e^{-} + \bar{\nu}_e$.

TABLE 8.15 Quark composition of decuplet baryons

Particle	Quark content	Charge (Q)	Baryon number (B)	Strangeness (S)
$\square^{++}$	*uuu*	$+\frac{2}{3}+\frac{2}{3}+\frac{2}{3}=+2$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	0 + 0 + 0 = 0
$\square^{+}$	*uud*	$+\frac{2}{3}+\frac{2}{3}-\frac{1}{3}=+1$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	0 + 0 + 0 = 0
$\square^{0}$	*udd*	$+\frac{2}{3}-\frac{1}{3}-\frac{1}{3}=0$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	0 + 0 + 0 = 0
$\square^{-}$	*ddd*	$-\frac{1}{3}-\frac{1}{3}-\frac{1}{3}=-1$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	0 + 0 + 0 = 0
$\square^{*+}$	*uus*	$+\frac{2}{3}+\frac{2}{3}-\frac{1}{3}=+1$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	0 + 0 – 1 = –1
		$+\frac{2}{3}-\frac{1}{3}-\frac{1}{3}=0$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	

□*0	*uds*			0 + 0 – 1 = –1
□*–	*dds*	$-\frac{1}{3}-\frac{1}{3}-\frac{1}{3}=-1$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	0 + 0 – 1 = –1
□*0	*uss*	$+\frac{2}{3}-\frac{1}{3}-\frac{1}{3}=0$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	0 – 1 – 1 = –2
□*–	*dss*	$-\frac{1}{3}-\frac{1}{3}-\frac{1}{3}=-1$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	0 – 1 – 1 = –2
□–	*sss*	$+\frac{2}{3}+\frac{2}{3}-\frac{1}{3}=+1$	$+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=+1$	–1 – 1 – 1 = –3

8.11.1 Observation of Quarks

The model faced several difficulties also alongside the successes achieved. For instance, the first problem was, "Do isolated quarks exist or these are always bound?". Other question was the apparent violation of Pauli exclusion principle. The first main problem was that though three quarks form a proton, it is impossible to separate them. Many careful and ingenious experiments have been performed with high-energy accelerators to liberate quarks, but without any success. Other searches for free quarks, for example in cosmic rays, stellar spectra, terrestrial and lunar rock samples, and deep ocean sediments, have also failed. Thus, *Hadron-spectroscopy* is different from the *atomic* and *nuclear spectroscopies*. Whereas the atomic and nuclear constituents could be isolated, quarks remain permanently confined in hadrons. In hadronic collisions it is energetically favourable to produce more hadrons rather than ejecting the quarks.

Indirect confirmation of the quark structure of proton has been obtained in deep inelastic lepton–nucleons collisions. Actually, what we see inside the proton depends on the scale on which one observes it. If a proton is scattered with radiation of a wavelength of a few times 10^{-15} m, say through 50 MeV lepton, it appears just a point-like object. It might contain all sorts of its constituents, but no structure is revealed at this scale. Boosting the energy of the probe to a few GeV, resolution gets increased and one finds the quark layer. High energy (E ~ a few GeV) beams of leptons, having de Broglie wavelength ($l = \square c/E \sim 10^{-18}$ m) << nucleon size (~10^{-15} m), reveal constituents (also called *partons*) of the proton. The results obtained from neutrino-nucleon scattering experiments, called *deep inelastic scattering*, reveal that the proton (and neutron) contains three point-like quasi-free quark constituents. These experiments

indeed confirmed that these quarks do have fractional charges, besides having spin ½. Further investigations also showed the existence of gluons, with which quarks are glued (bound) together to form hadrons.

8.11.2 Colour: A New Degree of Freedom

In order to see the second problem faced by the quark model, consider □ resonance spin-up state made up of three spin-up *u* quarks,

$$\Delta^{++}{}_{(I=3/2,\, I_3=3/2;\, S=3/2,\, S_3=3/2)} = u\uparrow u\uparrow u\uparrow$$

which is totally symmetric state under the interchange of any two quarks. This is in strict violation to the spin-statistics theorem and hence violating the Pauli exclusion principle. To overcome this violation, it was suggested that the antisymmetry of the baryon, being fermion, could be restored only if each quark is allowed an extra degree of freedom, called *colour*. These degrees have been named to make things a bit colourful, red, green and blue[16]. However, we do not need colour quantum number for the composite hadrons. That means hadrons are colourless (colour singlet).

16. This has nothing to do with ordinary visual colours.

Using the $SU(3)_{colour}$ group, where quark colours, red (r), blue (b) and green (g), belong to the fundamental multiplet of 3, and antiquark conjugate-colours $(\bar{r}, \bar{b}, \bar{g})$ belong to 3^*, the colour neutral states of the hadrons could be constructed as the decompositions given in Eq. (8.6) contain a singlet. Respective colour wave functions are:

$$|\,\text{Mesons}\,\rangle_{colour} = \frac{1}{\sqrt{3}}|\,\bar{r}r + \bar{b}b + \bar{g}g\rangle$$

$$|\,\text{Baryons}\,\rangle_{colour} = \frac{1}{\sqrt{6}}|\,rbg + bgr + grb - rgb - brg - gbr\rangle$$

To have agreement between theory and experiment for all these phenomena, we clearly need three colours. So each of these six quarks occurs in three varieties. Note that the colour assignment has nothing to do with the standard perception of colour.

8.12 RECENT DEVELOPMENTS

Particle physics has been an exciting and challenging frontier of physics. In the last few decades, lots of new developments have taken place in this subject, brief description of which is given below.

8.12.1 Quantum Field Theory of Interactions

Strong, electromagnetic and weak interactions are now described by quantum field theories. One may notice that all matter particles are fermions, whereas all the quanta of forces are bosons. In these theories, all elementary particles are described by their corresponding fields, which interact with each other guided by the so-called *principle of gauge invariance*. For instance, in the theory of electromagnetic interactions, known as *Quantum Electrodynamics* (QED), photon is a manifestation of electromagnetic field and electron is a quantum of the electron field, and photon, though itself electrically neutral, acts as the messengers of the electromagnetic interactions. QED was developed in 1948 independently by Schwinger, Feynman and Tomonaga. In this theory, electromagnetic interaction between two charges takes place through exchange of photon. This photon is virtual in the sense that energy ΔE can be borrowed to create the quantum, provided it is done only for a short time Δt, where $\Delta E \cdot \Delta t \sim \hbar$ (uncertainty principle). So an electron may be imagined as continually emitting and absorbing the photons. Sometimes these photons get absorbed by a nearby electron, thus giving rise to the electromagnetic force. This theory has achieved several excellent experimental confirmations: anomalous magnetic moment of leptons, Lamb shift say between $2s_{½}$ and $2p_{½}$ levels of hydrogen atom etc. It also predicts that the Coulomb law gets modified at very short distances due to effects like vacuum polarization (screening of electric charge).

In mid-sixties, electroweak field theory, called *quantum flavourdynamics* (QFD), was developed by Glashow, Wienberg and Salam in which *W*- and *Z*-bosons act as the mediator of the weak interactions, besides unifying the weak and electromagnetic interactions. Many predictions of this theory have been confirmed experimentally. For instance, it predicted the neutral current weak phenomena like $n + p \rightarrow n + p$, and possibility of parity violation in atomic forces, which have been observed. Similarly, for the strong interaction also, the observation of quasi-free nature of quarks inside the proton suggested a field theory of strong interactions should be possible.

Guided by the QED, a quantum field theory of strong interactions, known as the *Quantum Chromodynamics* (QCD), has been developed, in which the colour acts as the strong charge. Basically, QCD postulates the specific quanta, called *gluons*—mediating the strong interaction between the quarks, just as the photons mediate electromagnetic interaction. There are, however, some fundamental differences also. In contrast to the single electric charge, the strong colour charges are three in number, which belong to the fundamental multiplet of the SU(3) colour symmetry group. QED has only one photon, whereas the QCD requires 8 gluons.

Combining these three theories of strong, weak and electromagnetic interactions gives us the so-called 'Standard Model' of the fundamental interactions. Up to now all its predictions have been found to be in excellent agreement with experimental results. However, there are reasons to believe that the standard model needs to be extended further. Supersymmetry (SUSY) and Grand Unified Theories provide some of the possible generalizations of this model.

8.12.2 Unification of Fundamental Forces

Unification of fundamental forces has been a major challenge. However, every step at the unification programme has yielded some bonus. For instance, unification of electricity and magnetism, developed by Maxwell around 1860, provided optics as bonus. The fact that strong, electromagnetic and weak interactions are described by similar quantum field theories suggest their common origin. Theoretical physicists have already developed so-called **Grand Unified Theories** (GUTs) which unify all the three interactions and predict the violation of baryon number[17] (B). Such theories also predict the existence of heavy quanta (around 10^{15} GeV), called *lepto-quark*, which could make proton decay violating the baryon number conservation, for example, in the mode $p \rightarrow p^0 + e^+$. On the experimental side, careful investigations are being made to look for probable proton decay. At present, no such event has been observed so far and the only limit on its lifetime ($> 10^{33}$ years). Theoretically, proton instability acquires a deeper significance for explaining the origin and development of the universe. Particularly, baryon number violation (along

with possible C and CP violation) can play an important role in understanding the problem of matter–antimatter symmetry observed in the universe.

17. It also predicts violation of lepton number (L); however, it may conserve B–L quantum number.

At present much theoretical work is going in developing the *supersymmetry* (SUSY) and *supergravity* (SUGRA) and *superstring* field theories. As a last frontier, final unification of strong, weak and electromagnetic interactions with gravity is also being developed which is called *Theory of Everything* (TOE). However, it is far from completeness as it is suffering from serious difficulties, since gravity has not been understood at the quantum level around Planck mass scale $\sqrt{\hbar c/G_N} = 1.22 \times 10^{19}$ GeV. We may now hope to understand the creation of universe back up to Planck time ($\sqrt{\hbar G_N/c^5} = 0.54 \times 10^{-43}$ s) or Planck length ($\sqrt{\hbar G_N/c^3} = 1.61 \times 10^{-35}$ m), where quantum gravity takes over the domain. Thus, during the last few decades, close links between particle physics and cosmology have been established. Existence of dark matter and dark energy in the universe has provided new challenges to develop these links further. Status of unification theme is given in Table 8.16.

TABLE 8.16 Status of unification of fundamental interactions

Theory	Interactions unified	Bonus	Status
QED	Electricity and magnetism	Optics	Very successful
QFD	Electromagnetism and weak	Neutral currents	Successful
GUT	Strong, electromagnetism and weak	Proton decay	Promising
TOE	Strong, electromagnetism, weak and gravity	Cosmology	Probable

As a result of the successes of particle physics, we now believe that we have a good understanding of the natural phenomena ranging from microcosm (very small) to macrocosm (very large objects). For the first time, we are able to form a unified view of the cosmos. It has been learnt finally that nature's imagination is much richer than our vision. It may have many surprises in its store. For instance, quarks and leptons, today's fundamental particles, may have another layer of structure. Only future experiments would decide such issues.

NUMERICAL PROBLEMS

Section 8.3

Unsolved Problem

1. Prove that fine structure constant $a = \frac{1}{137}$.

Section 8.4

Unsolved Problems

1. Mean lifetime of □-baryon is 2.63 □ 10^{-10} s. Determine its decay width.

$$\left[\textbf{Hint. } \Gamma = \frac{\hbar}{t}\right]$$

2. Using time–energy uncertainty relation □E □t □ □, show that range (R) of a force mediated by quanta carrying rest mass (m) is given by $R = \frac{\hbar}{mc}$.

3. Calculate the range of weak forces using mass of W-boson in the range formula.

Section 8.5

Solved Problems

1. An antiproton comes to rest and annihilates with a proton. They produce p^+, p^- and p^0 of equal energy. What is the average kinetic energy of each pion in MeV?

Solution: The decay is

$p + \bar{p} \square p^0 + p^- + p^+$

Total energy of p and $\bar{p}$ system = 938 + 938 = 1876 MeV

Out of this, the energy consumed in creating 3 pions = 134.9 + 139.5 + 139.5 = 413.9 MeV

Energy left to be shared by 3 pions = 1876 – 413.9 = 1462.1 MeV

Therefore, average kinetic energy of each pion = $\frac{1462.1}{3}$ = 487.4 MeV

2. A particle X has two decay modes with partial decay rates r_1 s^{-1} and r_2 s^{-1}. What is the inherent uncertainty in mass of particle X?

Solution: The total decay rate of particle X is

$$l = r_1 + r_2$$

So, the mean lifetime of the particle is

$$\tau = \frac{1}{\lambda} = \frac{1}{r_1 + r_2}$$

Therefore, inherent uncertainty in the mass of the particle is given by the relation □ □ $t \cong \hbar$. Hence,

$$\Gamma \cong \frac{h}{\tau} = \hbar(r_1 + r_2)$$

Unsolved Problems

1. The □^{0}-hyperon decays to □$^0 + g$ with a mean lifetime of 7.4 □ 10^{-20} s. Estimate its width in keV. [**Ans.** 8.9 keV]

2. Find the distance covered by a muon having kinetic energy of 5 MeV in free space before it decays. Half-life of muon = 2.197 □ 10^{-6} s and its mass = 105.66 MeV. [**Ans.** 202.8 m]

3. Show that in natural units 1 kg = 5.63 □ 10^{26} GeV.

Section 8.6

Unsolved Problems

1. p^--mesons exist in three charged states +1, 0 and –1 (in units of proton's charge). Find out its strangeness. [**Ans.** 0]

2. □-baryons exist in three charged states –1, 0 and +1. Show that the □ has strangeness $S = -1$.

Section 8.7

Solved Problems

1. Explain why the following reactions cannot occur through strong interactions even if the kinetic energy of the incident proton is several GeV.

(i) $p + n$ □ $K^+ +$ □$^- +$ □$^+$

(ii) $p + n$ □ □$^0 +$ □$^+$

Solution: Since these reactions involve only strongly interacting particles, so all the conservation laws must hold good. If some law is violated, the reaction cannot take place in spite of the energy of the incident proton. In these reactions conservation of strangeness, isospin and I_3 are violated. Hence, these reactions cannot occur.

2. Which of the following reactions are possible?

(i) $p^+ + n \rightarrow \Delta^0 + K^+$

(ii) $p^+ + n \rightarrow K^0 + K^+$

(iii) $\bar{\nu}_e + p \rightarrow n + m^+$

(iv) $\bar{\nu}_e + p \rightarrow n + e^+$

Solution: The first reaction is

$$\pi^+ + n \rightarrow \Lambda^0 + K^+$$

Baryon number:	$0 \quad +1 \rightarrow +1 \quad 0$
Q:	$1 \quad 0 \rightarrow 0 \quad 1$
I_3:	$1 \quad -\frac{1}{2} \rightarrow 0 \quad \frac{1}{2}$
S:	$0 \quad 0 \rightarrow -1 \quad +1$

In this reaction baryon number, charge, strangeness, isospin, etc. are all conserved, so the reaction is possible through strong interaction.

The second reaction is

$$\pi^+ + n \rightarrow K^0 + K^+$$

Baryon number:	$0 \quad +1 \rightarrow 0 \quad 0$

In this reaction baryon number is not conserved, so the reaction is not possible through known interactions.

The third reaction is

$$\bar{\nu}_e + p \rightarrow n + \mu^+$$

Lepton number L_e:	$+1 \quad 0 \rightarrow 0 \quad 0$
Lepton number L_μ:	$0 \quad 0 \rightarrow 0 \quad -1$

For this reaction L_e and Lm both are not conserved, so this reaction is not possible through any known interaction.

The last reaction is

	$\bar{\nu}_e + p \rightarrow n + e^+$
Lepton number L_e:	$+1 \quad 0 \rightarrow 0 \quad +1$
Baryon number:	$0 \quad +1 \rightarrow 1 \quad 0$
Q:	$0 \quad 1 \rightarrow 0 \quad 1$

In this case, all conservation laws hold, so the reaction is possible through weak interactions.

3. Determine whether the following reactions are allowed or forbidden.

(i) $p + p \square K^+ + \square^+$

(ii) $p + p^- \square \square^0 + h^0$

(iii) $p + p \square p + p + p + {}^{\bar{p}}$

Solution: The first reaction is

$$p + p \square K^+ + \square^+$$

Charge: +1 +1 □ +1 +1 is conserved

Baryon number: +1 +1 □ 0 +1 is not conserved

Strangeness: 0 0 □ +1 –1 is conserved

Isotopic spin I_3: $\frac{1}{2}\frac{1}{2} \rightarrow \frac{1}{2} + 1$ is not conserved.

Hence, this reaction is not possible.

The second reaction is

$$p + p^- \square \square^0 + h^0$$

Charge: +1 –1 □ 0 0 is conserved

Baryon number: +1 0 □ +1 0 is conserved

Strangeness: 0 0 □ –1 0 is not conserved

Isotopic spin I_3: $\frac{1}{2}$ –1 □ 0 0 is not conserved.

Hence, this reaction is not possible through strong interactions.

The third reaction is

$$p + p \square p + p + p + {}^{\bar{p}}$$

Charge: +1 + 1 □ +1 + 1 +1 –1 is conserved

Baryon number: +1 +1 □ +1 +1 +1 –1 is conserved

Strangeness: 0 0 □ 0 0 0 0 is conserved

Isotopic spin I_3: $\frac{1}{2}\frac{1}{2} \rightarrow \frac{1}{2}\frac{1}{2}\frac{1}{2} - \frac{1}{2}$ is not conserved

Hence, this reaction is possible through strong interactions.

Unsolved Problems

1. Show that $K^+ \rightarrow p^+ + p^0$ violate parity conservation.
2. A particle of mass (M) at rest decays to two particles carrying masses m_1 and m_2. Determine the kinetic energy of decay products, using energy–momentum conservation laws.
3. State the conservation law, which forbids the following reactions:
   (i) $p + p \rightarrow g$
   (ii) $p \rightarrow p + e^-$
   (iii) $e^- \rightarrow g + n_e$
   (iv) $n \rightarrow p + p^-$
   (v) $p \rightarrow p + g$
   (vi) $K^- \rightarrow p^0 + e^-$
   (vii) $m^- \rightarrow e^- + g$
   (viii) $\Delta^0 \rightarrow \Delta^- + p$
4. State which of the following processes are allowed and which are forbidden. Give reasons.
   (i) $\Delta^0 \rightarrow \Delta^0 + n$
   (ii) $m^- + p \rightarrow \Delta^0 + nm$
   (iii) $\Delta^0 \rightarrow \Delta^0 + m^+ + nm$
   (iv) $K^- + d \rightarrow p^+ + \Delta^-$
   (v) $K^+ \rightarrow p^0 + e^+ + n_e$

   [**Ans.** (i) allowed, (ii) allowed, (iii) forbidden by energy conservation $mm > m_{\Delta} - m_{\Delta}$, (iv) forbidden by baryon conservation, (v) allowed].
5. Why the following reactions are allowed?
   (i) $p + p \rightarrow p + \Lambda + \Sigma^+ + \Xi^0$

(ii) $p + \bar{p} \rightarrow \Sigma^0 + \bar{\Lambda}$

(iii) $g + n \rightarrow n + p^0$

(iv) $p^- + p \rightarrow \Sigma^0 + K^0$

(v) $p^+ + p \rightarrow \Sigma^0 + K^+ + K^+$

(vi) $p^- + p \rightarrow \Sigma + K^+ + K^-$

6. Why the following reactions are forbidden through strong interactions?

(i) $p + p \rightarrow \Sigma^+ + p$

(ii) $\bar{p} + p \rightarrow \Sigma + \Sigma$

(iii) $p^- + p \rightarrow \Sigma^+ + K^-$

(iv) $K^+ + p \rightarrow \Sigma^+ + p^+$

7. State the conservation law, which forbids the following decays:

(i) $p + \bar{p} \rightarrow g$

(ii) $n \rightarrow p + e^-$

(iii) $e^- \rightarrow g + n_e$

(iv) $n \rightarrow p + p^-$

(v) $p \rightarrow p + g$

(vi) $K^- \rightarrow p^0 + e^-$

(vii) $m^- \rightarrow e^- + g$

(viii) $\Sigma^0 \rightarrow \Sigma^- + p$

8. The Σ^0-hyperon is seen to decay as $\Sigma^0 \rightarrow \Lambda + p^0$ but not as $\Sigma^0 \rightarrow n + p^0$ even though neutron (n) is lighter than Λ-hyperon. Explain why.

Section 8.9

Unsolved Problems

1. Determine the charged state of a resonant baryon having isospin $I = \frac{3}{2}$ and zero strangeness. [**Ans.** ++, +, 0, –]

2. Assume mass operator for isomultiplet to be given by

$$M = a + bI_3$$

where a and b are unknown parameters and I_3 denotes third component of isospin, show that decuplet baryons satisfy the following equal spacing relation:

$$m_{\Delta^{++}} - m_{\Delta^{+}} = m_{\Delta^{+}} - m_{\Delta^{0}} = m_{\Delta^{0}} - m_{\Delta^{-}}$$

3. Show that a set of Pauli matrices s_i satisfy the following:

$s_i s_j - s_j s_i = 2is_k$

where $i, j, k = 1, 2, 3$ cyclically

4. □$^{*+}$ is an hyperon observed to have mass 1.385 GeV and decay width 35 MeV. Its dominant decay mode is □$^{*+}$ □ □ + p^{+}. Is this decay strong, electromagnetic or weak? Give reasons for your answer.

Section 8.10

Unsolved Problems

1. Λ_c^+-baryon has $I = 0$, $S = 0$ and $C = 1$. What is its quark content? [**Ans.** *udc*]

2. What values of electric charge are possible for a baryon in quark model?

[**Ans.** ++, +, 0, –]

3. What values of electric charge are possible for a meson in quark model? [**Ans.** +, 0, –]

4. Given that meson masses are expressed in terms of its constituent quark masses (m_i) and spin (s_i)

$$m_1 + m_2 + a\frac{\vec{s}_1 \cdot \vec{s}_2}{m_1 m_2}$$

Determine the masses of pseudoscalar and vector mesons, taking $m_d = m_u = 320$ MeV, $mt = 480$ MeV and $\frac{a}{m^2 u} = 640$ MeV.

REVIEW QUESTIONS

Short Answer Type

1. What are baryons?
2. What is the difference between *p*-mesons and *K*-mesons?
3. What are baryon and lepton conservation laws?
4. Assign the isospin quantum numbers to nucleonic doublet and pionic triplet.
5. Give the quark structure of a proton, neutron and meson.
6. Explain the concept of charge conjugation.
7. Name different types of quarks.
8. Explain the concept of isospin.
9. What are the quarks make-ups of neutron and proton?
10. What is CP violation?
11. What are quarks? Outline the basic assumptions and properties of quarks.
12. Classify the various types of interactions in relation to elementary particles.
13. Give the charges of any four types of quarks.
14. Explain why the decay

$$\square^0 \square \square^0 + g$$

is observed, but not

$$\square^0 \square r + p \text{ or } \square^0 \square n + p^0$$

15. What are resonances?
16. State CPT theorem.
17. What do you understand by isospin?
18. How do you differentiate between leptons and baryons?
19. What are leptons?
20. What is strangeness?
21. What are quarks? Are they coloured?
22. *m*-meson belongs to:
    (i) Leptons
    (ii) Baryons
    (iii) Photons
    (iv) Pions
23. Meson is a bound state of

(i) 3 quarks
(ii) 2 quarks
(iii) 5 quarks
(iv) None of these

24. Write three properties of quarks.

25. Mesons are:

(i) Leptons
(ii) Fermions
(iii) Bosons
(iv) None of the above

Long Answer Type

1. What are quarks? Explain quark flavours.

2. Explain which of the following reactions are allowed and forbidden under the conservation of strangeness, conservation of baryon number and conservation of charge.

(i) $p^+ + n \rightarrow \Lambda^0 + K^+$

(ii) $p^+ + p \rightarrow K^0 + K^+$

(iii) $p^- + p \rightarrow \Lambda^0 + K^0$

(iv) $p^- + p \rightarrow p^0 + K^0 + \Sigma^+$

3. Discuss the classification of particles on the basis of isospin.

4. What are fundamental particles? Discuss the Gell–Mann–Nishijima scheme of classification of particles.

5. Discuss the quark structure of nucleons.

6. Write down the four basic interactions in nature giving their relative strengths and also name the exchange particles responsible for them.

7. Explain the concept of associated production.

8. Discuss the concept of associated production and give the decay modes of some of the strange particles.

9. Construct quark structure of a nucleon and a pion.

10. Compare the properties of leptons and baryons.

11. Explain the concept of strangeness and associated production. Discuss the Gell–Mann–Nishijima relation.

12. Explain in detail the quark model and explain the various types of quarks (flavours) along with their properties.

13. Explain the terms isotopic spin and strangeness. In what respect are they important in the classification of elementary particles?

14. Write a brief note on different types of interactions which the elementary particles can undergo. Give examples.

15. Give reasons to explain the associated production of strange particles and charmed particles.

16. What are quarks? Give the elementary theory of structure of a few hadrons on the basis of quark model.

17. Write notes on:

(i) Isospin quantum number

(ii) Strange particles and their decay modes

18. What are quarks? Give the qualitative description of quark model.

19. Give the quark model of

(i) mesons

(ii) proton and antiproton

(iii) neutron and antineutron

20. Describe briefly the following intrinsic quantum numbers in connection with elementary particles:

(i) Lepton number

(ii) Baryon number

(iii) Isospin

(iv) Hypercharge

(v) Strangeness

(vi) Decay process of mesons.

What conservation laws govern them? Give at least one example in support of each conservation law.

21. Discuss the quantum numbers associated with elementary particles. Give the corresponding conservation laws. Give at least one example in support of each conservation law.

22. Discuss the various types of interactions between elementary particles giving their characteristic coupling constant and lifetimes. What are the interaction carriers in electromagnetic and strong interactions between nuclei?

23. Discuss the following intrinsic quantum numbers associated with elementary particles:

(i) Charge number

(ii) Baryon number

(iii) Lepton number

(iv) Isospin quantum number

(v) Hypercharge

(vi) Strangeness

(vii) Decay process of mesons

Explain with example.

24. What are quarks? Explain the quark model. Also give two factors which do not support the existence of quarks.

25. Classify the various types of elementary particles in reference to their lepton number, baryon number and isospin.

26. What is charge conjugation? Explain with example.

27. What is meant by colour of a quark? Give the colours associated with quarks.

28. Discuss the quark model and explain how mesons and baryons are formed using quarks.

29. Explain:

(i) Isospin

(ii) Gell–Mann–Nishijima scheme

30. Define parity, charge conjugation and time reversal. State CPT theorem.

31. A positron and an electron having negligible kinetic energies annihilate each other to produce two photons. Calculate the frequencies of the two photons.

32. Which of the following can occur? State conservation laws violated by others.

(i) $p + p \rightarrow n + p + p^{+}$

(ii) $p + p \rightarrow p + \square^{0} + \square^{+}$

(iii) $e^{+} + e^{-} \rightarrow m^{+} + p^{-}$

33. What are leptons? Name any three leptons and their antiparticles. Briefly discuss the properties of leptons. Discuss the statement, *m*-meson is also a lepton.

34. What is charge conjugation?
35. Discuss in detail the strange particles.
36. What is quark model? Is there any evidence in its support?
37. Discuss the decay modes of mesons.
38. What are hyperons and leptons? Give their decay modes.
39. Write notes on:
(i) Isospin
(ii) Hypercharge
(iii) *K*-mesons
40. What are elementary particles? Give a brief account of the discovery and properties of elementary particles.